Highlights

Beyond forecasting: an empirically grounded agent-based model of how household decisions, policy and infrastructure shape the Dutch heat transition

Naud Loomans, Floor Alkemade, Lisanne Havinga

- Agent-based model of heating choices for the full Dutch housing stock
- Gas declines robustly, all-electric heat pumps are the main substitution
- Hybrid heat pumps are the fallback, district heating needs strong policy
- Feedback loops, policy measures, and infrastructure shape the pathways
- Bottom-up models map pathways and barriers rather than a single forecast

Beyond forecasting: an empirically grounded agent-based model of how household decisions, policy and infrastructure shape the Dutch heat transition

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Abstract

The transition to sustainable heating in Dutch homes depends on the interplay between the decisions of millions of owner-occupiers, landlords, and housing associations, with infrastructure planning, and policy. We investigate how this interplay shapes transition pathways and what an empirically grounded bottom-up model can contribute when data is limited, and future decision-making is inherently uncertain. Two conclusions stand out. First, the decline of natural gas is robust, but which technologies replace it depends on behavior, policies, and infrastructure. Second, such a model is most useful not for forecasting a single pathway but for identifying no-regret developments, the barriers to desired outcomes, and what it takes to align household decisions with policy goals. Our agent-based model of the full Dutch housing stock combines decision models for owner-occupiers, landlords and housing associations with economic learning, social influence, grid congestion and district-heating expansion. We run an ensemble of plausible decision weights through scenarios, consumer segments and decision structures. Under current rules, all-electric heat pumps are the main alternative to gas. Hybrid heat pumps are the main fallback, whereas district heating expands only with strong policy support. Because consumer segments respond differently, policy can be tailored to the groups it needs to reach.

Keywords: agent-based modeling, transition pathways, residential heating, technology adoption, uncertainty

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1. Introduction

Energy system models and scenarios are the predominant tools used by policymakers to plan and govern the energy transition. They strongly shape perceptions of feasible pathways, costs and timelines, and therefore influence investment and regulatory decisions. Most of these models, however, determine pathways by cost optimization or stylized simulation, and are not designed to ask how individual actors actually decide or what is realistic. Real-world transitions deviate substantially from cost-optimal pathways (Trutnevyte, 2016), because they are shaped by heterogeneous actors, social influence, institutions and infrastructure, and by path dependence and feedback loops that amplify early choices (Barazza and Strachan, 2021; Mercure et al., 2019; Fouquet, 2016). Assessing what is feasible therefore requires combining techno-economic with socio-technical and political perspectives (Cherp et al., 2018; Geels et al., 2016).

This gap is particularly large for demand-side transitions such as residential heating. Here, the transition is not decided by a few large actors but by millions of owner-occupiers, landlords and housing associations, each deciding at their own moment, typically when a heating system fails or a renovation is planned. These decisions depend not only on costs, but also on attitudes, social influence, the dwelling and its ownership, and on infrastructure. The infrastructure itself, such as electricity grids and heat networks, is planned by other actors who again depend on these household decisions for utilization. (Rao et al., 2025; Hesselink and Chappin, 2019).

Socio-technical energy transition (STET) models have been proposed to bridge this gap, by combining techno-economic detail with explicit representations of actors, behavior and institutions (Li et al., 2015; Hirt et al., 2020; Trutnevyte et al., 2019). Agent-based models in particular have been used to simulate the adoption of energy technologies from individual decisions (Rai and Robinson, 2015; Kiesling et al., 2012; Moglia et al., 2017). Many of these models are stylized or local. Those that are empirically grounded face a fundamental data problem, the behavioral parameters that drive the simulated dynamics cannot be identified from the short and noisy adoption records that are available (Holtz et al., 2015; McDowall and Geels, 2017; Köhler et al., 2018).

The contribution of this paper lies in bridging these two gaps. We develop an empirically grounded agent-based model of the Dutch residential heat transition and use it to explore transition pathways. First, whereas most

STET models are stylized or local, ours covers the full Dutch housing stock with techno-economic detail per dwelling, heterogeneous actors and feedback loops. It combines a theory-of-planned-behavior decision model for owner-occupiers (Okur et al., 2024) with cost-based decisions for landlords and housing associations, and represents economic learning, social influence, grid congestion and the expansion of district-heating networks, so that its pathways can be compared with national transition scenarios. Second, whereas empirically grounded models are usually calibrated to a single parameterization, we keep the empirical grounding while exploring the uncertainty that calibration cannot reduce. Only two years of consistent adoption data are available, from which calibration cannot pin down an decisive set of decision weights. Rather than choosing one, we retain this ensemble of plausible weight sets and carry it through a broad set of scenarios, consumer segments, sensitivity analyses and an alternative decision structure.

We address two questions. First, which features of the transition pathway are robust across plausible behavior, and which depend on behavior, policy and infrastructure? Second, what can an empirically grounded bottom-up model contribute to transition planning when data are limited and future decision-making is inherently uncertain? The Dutch case suits these questions. More than eight in ten homes still use natural gas, and several low-carbon alternatives compete, each relying on different infrastructure.

Section 2 introduces the Dutch residential heat transition and Section 3 the transition dynamics represented in the model. Section 4 describes the model, Section 5 the analysis design and Section 6 the calibration. Section 7 presents the results, which are discussed in Section 8; Section 9 concludes.

2. Residential heat transition in the Netherlands

The residential heat transition in the Netherlands is a timely case, as it stands at the threshold of a new phase that requires decisive action in the coming years. The current heating regime is based on natural gas. In 2025, 76 % of homes had an individual gas boiler and another 5 % collective gas heating (CBS, 2026a). This is the most gas-dominated residential heating regime in Europe. For reference, the United Kingdom, Germany, Belgium and Italy also have significant shares of natural gas based heating, while the highest shares of district heating are found in Scandinavian and Eastern European countries (Bertelsen and Mathiesen, 2020).

Three main alternatives compete to replace the gas boiler: the hybrid heat pump, the all-electric heat pump and district heating. Where each alternative has its own dynamics, and prefered solutions differ per stakeholder, dwelling, and local context (Loomans and Alkemade, 2024). Owner-occupiers have rapidly adopted (hybrid) heat pumps in recent years, but this pathway is now entering a bottleneck due to grid congestion. Studies point to lower societal costs when more district heating is part of the mix (van Polen et al., 2026; PBL, 2020), but for individual consumers its costs are high (Salite et al., 2024). The choice is made by owner-occupiers, private landlords, housing associations and owners' associations, while municipalities, grid operators and district-heating operators plan the infrastructure on which each alternative depends.

The Dutch transition scenarios and policy plans show how open the route still is. Depending on the scenario of the Dutch grid operators, hybrid heat pumps heat between 10 and 60 % of homes in 2050, electric heat pumps between 25 and 55 % and district heating between 10 and 30 % (Figure 1, Netbeheer Nederland, 2025). Municipal heat plans reflect this uncertainty, with about 20 % of dwellings planned on district heating, and 65 % with several options kept open or no plan yet (Molen et al., 2023). These differences reflect not only the uncertainty about the pathway, but also the different roles of these organizations. Households are adopting heat pumps, in line with the current regime of individual solutions, whereas municipalities have more influence on investments in district heating and see it as a way to reach their climate goals with less dependence on individual household decisions. Which pathway materializes depends on the decisions of millions of owners and their interactions, which aggregate scenario models represent only implicitly.

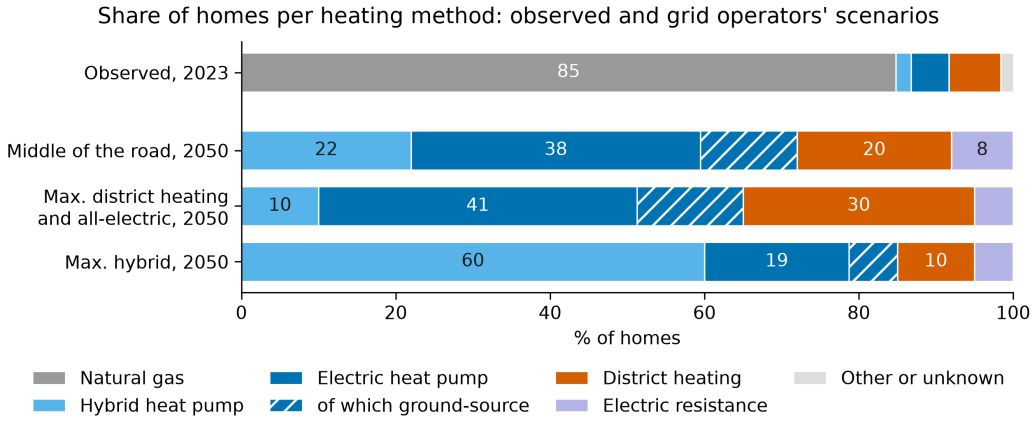


Figure 1: Share of homes per heating method, observed in 2023 (CBS, 2025) and in 2050 in three scenarios of the Dutch grid operators (Netbeheer Nederland, 2025). The scenarios are labelled by the heating choice they represent: the middle of the road (Koersvaste Middenweg), the maximum of district heating and all-electric heat pumps (Eigen Vermogen), and the maximum of hybrid heat pumps (Horizon Aanvoer). Natural gas includes individual and collective heating

3. Transition dynamics in the heat transition

Socio-technical transitions emerge from the interplay of many actors, technologies, institutions, and infrastructures (Geels, 2002, 2005). The literature describes a wide range of dynamics in this interplay, among them the S-shaped diffusion of innovations (Rogers, 2003a; Bass, 1969), threshold models of technological transitions (Zeppini et al., 2014), interactions between niches and incumbent regimes (Geels and Schot, 2007), feedback between policy, technology, firms, users and the wider public (Geels and Ayoub, 2023), and cascades across systems (Geels and Andersen, 2026). A model of a specific transition cannot represent all of them, and should focus on those most relevant to the case. The socio-technical energy transition (STET) perspective characterizes such models by techno-economic detail, explicit actor heterogeneity and transition pathway dynamics (Li et al., 2015). Based on the characteristics of the Dutch heat transition, we translate these into three dynamics: actor heterogeneity in decision-making, feedback loops in economic and social learning, and dependencies on infrastructure shaped by economies of scale. Techno-economic detail enters all three, through the costs for actors, the learning curves of the technologies, and the costs and developments of networks. These dynamics are typical of transitions in which

many actors make long-term investments that depend on shared infrastructure, and are usually simplified in energy models and scenarios.

Heterogeneous decisions and decision rules.. Adoption models of sustainable technologies typically combine economic, psychological and social factors rather than a single threshold such as cost parity (Rao et al., 2025). In the heat transition, these factors differ not only between households, but also in how decisions are made. Owner-occupiers weigh costs against a range of psychological, social, financial and practical factors (Jager et al., 2025). Landlords decide mainly on costs, but face a split incentive, as tenants usually pay the energy bill (Aydin et al., 2026). Housing associations invest mostly in steps, as part of planned maintenance, guided by sectoral targets and by performance agreements with their municipality, and are key partners in the neighborhood-oriented expansion of heat networks (van der Bent, 2022). Lastly, in apartment blocks with collective heating, owners have to decide collectively through the owners' association. Furthermore, from a techno-economic perspective, the suitable options differ per dwelling type, installed system, and insulation level. These differences in groups and consumer segments means that niches are created, for whom different heating solutions are most suited. These niches can serve as incubators where learning and feedback loops can be triggered and increase the competitive advantage of one solution over another.

Reinforcing feedback loops.. Two reinforcing loops connect individual decisions over time. In economic learning, every installed unit lowers the cost of the next through learning-by-doing (Arthur, 1989). Here, the alternatives differ (Loomans and Alkemade, 2024). Heat pumps are a granular technology that households adopt at their own moment, and granular technologies tend to learn fast (Wilson et al., 2020). District heating relies on large projects with economies of scale, which are prone to delays and cost overruns (Flyvbjerg and Gardner, 2023). In social learning, households are influenced by what their peers do, as in threshold and network models of diffusion (Granovetter, 1978; Watts, 2002; Centola and Macy, 2007). Such loops can produce non-linear, self-reinforcing adoption and, when all parts of the system align and thresholds are crossed, the tipping dynamics described in the transitions literature (Geels and Ayoub, 2023; Lenton et al., 2022). Whether they accelerate an alternative or strengthen the incumbent, however, depends on the state of the system. Social learning that follows

the prevailing norm reinforces the incumbent, whereas a committed minority in closely connected groups can overturn an established convention once it reaches a critical mass (Centola and Macy, 2007; Centola et al., 2018).

Infrastructure, policy and coordination.. Transitions are shaped by the inertia of existing infrastructure, rules and institutions (Geels, 2005; Verbong and Geels, 2007). In the heat transition, every option depends on a network: the electricity grid, the gas grid or a heat network. Because the costs of these networks are largely fixed, the cost per household falls as more households connect in an area, and rises for those who remain when others leave. These increasing returns create path dependence and lock-in (Arthur, 1989; Unruh, 2000), and make a single system per area the cheapest. This, however, conflicts with freedom of choice, and requires rapid action and strong coordination, as otherwise owner-occupiers invest in whichever solution fits their dwelling and preferences best. The networks are planned by other actors than the households who use them, and how their costs are passed on to users is a policy choice. Housing associations can act as launching customers for heat networks (van der Bent, 2022), whereas without a connection obligation, district heating carries a demand risk when individual alternatives are preferred (Herreras Martinez et al., 2023). In the Netherlands, municipalities take a central, orchestrating role in this coordination (Herreras Martínez et al., 2022). The heat transition is also coupled to other systems, as electrified heating and mobility meet on the same electricity grid (Loomans and Alkemade, 2024; Rinaldi et al., 2021).

These dynamics are also where the main uncertainty lies. Costs and technical solutions are relatively well documented, but how actors weigh them, how strongly social learning acts, and how infrastructure and policy develop are not. Because the dynamics interact and reinforce themselves, differences in them can lead to substantially different pathways. Transition models are therefore better used to explore the range of pathways that plausible dynamics produce than to predict one (Holtz et al., 2015; Moallemi et al., 2019; Walker et al., 2013). This exploration is the core of our analysis. Table 1 structures the remainder of this paper by summarizing, for each dynamic, the characteristic of the case that makes it important, how it is represented in the model, how its uncertainty is treated, and where its effects are analyzed in the results.

Table 1: The three transition dynamics, with the characteristic of the Dutch case that makes them central, their representation in the model, the treatment of their uncertainty, and the sections in which they are analyzed.

Dynamic	**Characteristic of the case**	**Representation in the model**	**Treatment of uncertainty**	**Section**
Heterogeneous decisions and decision rules	Millions of owner-occupiers, landlords and housing associations, who weigh costs, attitudes and norms differently	Theory of planned behavior for owner-occupiers, cost-based choice for landlords and housing associations	Calibrated to observed adoption, retaining an ensemble of plausible decision weights, and an alternative cost perception	6.1, 7.3, 7.5
Reinforcing feedback loops	Competing alternatives that become cheaper and more familiar as more households adopt them	Learning curves per technology and peer influence on the social norm	Sensitivity analysis and scenarios with weak and strong loops	7.1, 7.2
Infrastructure, policy and coordination	Each alternative depends on a network planned by other actors. Policies shape cost allocation, connection obligations and the impact of congestion	Grid congestion, heat-network expansion, connection obligations and costs, and the strategy of housing associations	Policy and infrastructure scenarios	7.4

4. Modelling transition dynamics

We capture the dynamics of Table 1 with an agent-based model. A method suited to heterogeneous actors who influence each other and their environment, and widely used to model energy transitions (Hansen et al., 2019; Li et al., 2015), technology adoption (Moglia et al., 2017; Du et al., 2022) and infrastructure decisions (Chappin and Dijkema, 2010). Compared to the adoption of electric vehicles or solar panels, the residential heat transition has received less attention (Hansen et al., 2019) and differs in two ways. Households choose between several competing alternatives rather than making a yes-or-no decision, a competition often missing in adoption models (Hesselink and Chappin, 2019). Second, the choice depends strongly on

the dwelling and on contextual factors such as the construction of a heat network. Existing models mostly address heat pumps and insulation by individual owner-occupiers (Rao et al., 2025), with notable exceptions for owner associations and thermal energy communities (Nava Guerrero et al., 2019; Fouladvand et al., 2020) and district heating (Busch et al., 2017; Akhatova et al., 2022). Our model combines these elements for the full housing stock.

4.1. Model description

The model simulates every dwelling in the Netherlands in annual time steps. Each year, the owners of dwellings whose heating system reaches the end of its life, or who face an opportunity to replace it, decide on their heating method. Afterwards the district-heating operator and the grid operator respond (Figure 2, Table 2). Neighborhoods are the spatial unit at which networks are planned and grid capacity is assessed. With these actors, all above-mentioned dynamics can be modelled. Heterogeneous dwelling owners with a high level of detail on ownership and dwelling data based on public databases, economic learning based on cumulative installed capacity per heating method, social learning through social networks, infrastructural detail through the electricity and heat grid operators, and policy measures by means of scenarios. Policy enters through the scenarios, as rules, regulations and subsidies, rather than as an endogenous feedback, because quantitative evidence on how Dutch heat policy responds to adoption is lacking. The model is documented in full following the ODD protocol in Appendix A.

Each of the actors is further described in the following section.

4.1.1. Dwellings and costs

Each dwelling has its construction year, size and type from the Dutch building register (Kadaster, 2022), and a heating method, in-house heat distribution system and insulation level drawn from neighborhood statistics. Heating options are compared by their equivalent annual cost (EAC), which puts options with different investments, running costs and lifetimes on an equal footing (Hastings, 2021). The EAC includes the insulation and low-temperature in-house heat distribution that an electric heat pump requires. These raise the investment but lower the heat demand. Heat demand, electricity demand and insulation costs per dwelling type, construction year and energy label are taken from the Vesta MAIS model (PBL, 2017; van der Molen et al., 2021). Independently of heating decisions, dwellings are renovated on average once every 25 years, which improves insulation and heat

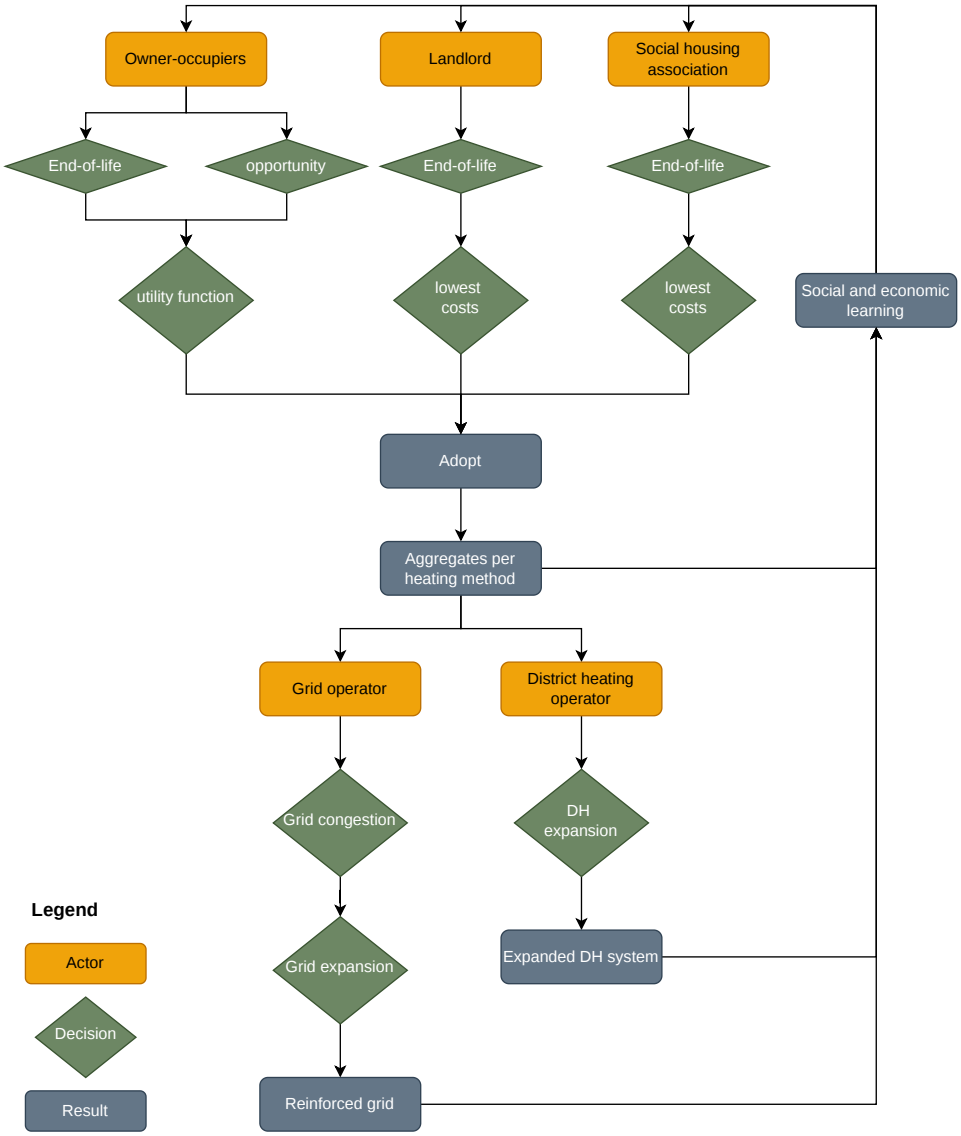


Figure 2: Flowchart of model dynamics.

Table 2: Description of agent populations in the model (M: million).

Agent	**Number**	**Behavior**
Dwellings	∼8 M	Each single dwelling in the Netherlands is simulated. Its characteristics are determined from neighborhood statistics, and they are attributed to decision-making agents based on their ownership.
Owner-occupiers	∼4.6 M	Each owner-occupied dwelling is assigned a household which owns the dwelling. On an annual basis each household makes an adoption decision on heating source and insulation based on a utility function combining costs with attitude and social norms, and the possibility of each option.
Social housing association	1	The social housing company has the goal to get a climate-neutral housing stock by 2050. To do this they can invest in insulation and sustainable heating methods. They assess groups of households on a street or block level (postal code 6), making annual adoption decisions for all included houses at once.
Landlords	∼1.2 M	All dwellings owned by landlords are assigned a landlord. Like owner-occupiers, landlords make annual adoption decisions. However, they are primarily focused on economic revenue instead of social and psychological norms and values.
Grid operator	1	The grid operator decides where to reinforce the electricity grid, making decisions per neighborhood. The impact of this depends on policy scenarios regarding grid congestion.
District heating company	1	District heating companies make investment decisions on where to build district heating grids per neighborhood. They have two potential strategies for expansion, based on heat demand density or on policy plans.

distribution at random and so changes the cost comparison over time. After all owners have decided, the investment costs of each technology fall with its cumulative installations along a learning curve (Wright's law). Energy prices follow the Dutch Climate and Energy Outlook (PBL, TNO, CBS & RIVM, 2026) translated into consumer tariffs including taxes, district heat is priced at the regulated tariffs for both energy and connection costs (ACM), and investment subsidies follow the Dutch schemes. These assumptions, the

spread of equipment lifetimes, and the exogenous renovation of the stock are detailed in Appendix A.2.

4.1.2. Owner-occupiers

The 4.6 million owner-occupied dwellings (57 % of the stock) are each represented by a household that decides on its heating method. Household adoption models of sustainable heating are varied. The theory of planned behavior (TPB) is the most used framework (Sopha et al., 2011, 2013; Okur et al., 2024), but many models use a utility function without a theoretical basis (Rao et al., 2025). Commonly included factors are economic utility, in the form of investment costs, discounted annual costs or payback times (e.g. Snape et al., 2015; Meles and Ryan, 2022; Michelsen and Madlener, 2013), often combined with detailed modelling of a heterogeneous building stock (e.g. Lee et al., 2014); attitudes towards sustainability, ease of use or novelty (e.g. Snape et al., 2015; Meles and Ryan, 2022); and social influence, through the number of adopters in a network or through explicit peer networks (e.g. Snape et al., 2015; Nava Guerrero et al., 2019; Meles and Ryan, 2022). A review of agent-based models of household energy decisions shows that models of investment decisions grounded in a behavioral analysis include more social determinants, in line with the role of peers in such decisions, but that almost half of the models do not report why they include the determinants they do (Rietkerk et al., 2025). We therefore take the determinants from an empirical study of Dutch owner-occupiers' heating choices (Okur et al., 2024).

Models also differ in the structure of the decision. Most utility-based models use a single stage, in which a household adopts once a threshold is reached (e.g. Snape et al., 2015; Meles and Ryan, 2022). Multi-stage models acknowledge that heating decisions are complex (Ebrahimigharehbaghi et al., 2019). Hecher et al. (2017) distinguish a trigger, a selection and a decision stage, with triggers that are problem-based, opportunity-based or a new building. A problem-based trigger, when the current system fails, mostly leads to like-for-like replacement, whereas an opportunity, such as a subsidy program or a planned renovation, more often leads to a switch (Broers et al., 2019). Because heating systems are replaced only at such moments, the pace of the transition is bounded by their lifetime, which creates inertia in the incumbent regime. Even when an alternative becomes attractive, it is adopted only as existing systems are replaced. We therefore trigger decisions both at the end of a system's life and at opportunities before it, after which the

household selects the option with the highest utility. For this utility we use a relatively simple TPB function following Okur et al. (2024), as the focus of this paper is on the interactions between actors rather than on the detail of the individual decision.

The attitude of a household towards each heating method is fixed at initialization. The subjective norm reflects social influence. Its basis is the share of peers in the household's network using a method. This share is amplified by the method's salience σ_{hm} (Equation 1), which can be seen as its visibility or a per-method measure of how much attention the method draws. It combines two sources of visibility (Equation A.1): novelty, which makes a method stand out while it is still rare, and prevalence, which makes a common method the norm. The faster a method's cumulative installations grow, the more weight novelty receives, otherwise salience follows prevalence. Adoption of an emerging, rapidly spreading method by peers therefore weighs more than that of an established one, but only through the household's own network. The strength of this channel is scaled by a social-learning rate that is varied in the scenarios. Perceived behavioral control (PBC) combines affordability, one minus the EAC normalized on a logarithmic scale across all options of the year, and the effort of switching (Equation 2). Attitude, subjective norm and PBC combine into an intention (Equation 3), and intention and PBC into a utility (Equation 4). Of the options available to the dwelling (for instance, district heating only where a network exists), the household chooses the one with the highest utility plus a random term, which represents everything the utility does not capture (Equation 5). Here ε_{hm} is an independent standard Gumbel draw per option and decision, and λ its scale.

$$SN_{hm} = \min(1,\ s_{hm} \cdot (1 + \sigma_{hm})) \tag{1}$$

$$PBC_{hm} = \frac{(1 - EAC_{hm,norm}) \cdot w_{costs} + (1 - Effort_{hm}) \cdot w_{effort}}{w_{costs} + w_{effort}} \tag{2}$$

$$Intention_{hm} = \frac{Att_{hm} \cdot w_{Att} + SN_{hm,n} \cdot w_{SN} + PBC_{hm,n} \cdot w_{PBC,int}}{w_{Att} + w_{SN} + w_{PBC,int}} \tag{3}$$

$$Utility_{hm} = \frac{Intention_{hm} \cdot w_{intention} + PBC_{hm} \cdot w_{PBC,util}}{w_{intention} + w_{PBC,util}} \tag{4}$$

$$\text{Selected heating method} = \underset{hm \in HM}{\arg\max} \left(Utility_{hm} + \lambda \, \varepsilon_{hm} \right) \tag{5}$$

Attitudes are drawn from distributions based on Okur et al. (2024) and Broers et al. (2019), with correlated parameters drawn by Cholesky decomposition (Edlin et al., 2015), and each household is connected to a small-world peer network in which it links to owner-occupiers with a similar attitude in its municipality, similar to the method used by Loomans et al. (2026).

4.1.3. Other actors

Housing associations and landlords own 29 and 15 % of the stock. Their actual decisions involve tenants, local governments and, for private landlords, a wide variety of motives (Filippidou et al., 2017; Hoppe, 2012; Lennartz et al., 2019; März, 2018). As data to model this are lacking, both are represented as cost-minimizing actors that choose the option with the lowest EAC when a system needs replacement. Although tenants usually pay the energy bill, we use the full EAC, as lower energy costs can be reflected in rents (Chegut et al., 2016). Housing associations decide for all their dwellings in a block at once, and can alternatively follow the municipal heat plans (Molen et al., 2023), in which they can act as launching customers for heat networks.

The district-heating operator builds heat networks per neighborhood, either following municipal plans or where the heat demand of the neighborhood's dwellings exceeds a break-even density of 600 GJ/ha. A network becomes operational after a construction time of five years or for density-based networks the completion year is drawn at random up to 2045, as not all networks can be built at once. Once a network exists, all dwellings in the neighborhood can connect. In practice, public acceptance, policy support and business models are also essential to such expansion (Bertelsen et al., 2021; Salite et al., 2024).

The grid operator computes the maximum electricity demand per neighborhood after all adoption decisions, including that of electric vehicles, which follow an S-shaped adoption curve, and reinforces the grid where the capacity is exceeded. Until then, the grid can constrain the adoption of electric heat pumps depending on scenario assumptions.

5. Analysis design

The dynamics of Section 3 are hard to calibrate and validate. Data on them are scarce, and fitting the model to current adoption would extrapolate

current trends, while we are interested precisely in the deviations from those trends. We therefore use an exploratory design that treats each dynamic according to what can be known about it (Table 1).

The decisions of owner-occupiers are governed by the decision weights in Equations 2–4. How strongly attitude, peers, affordability, and effort count in the choice of a heating method. The weights determine current day adoption and can therefore be estimated using calibration. We calibrate them together with the strength of the random term in the choice, against observed adoption from 2022-2024. As the short data series cannot single out one set of weights, we retain all sets with a good enough fit as an ensemble rather than a single best fit. The strength of the feedback loops, in contrast, shows mainly in how adoption develops beyond the observed years, and is explored in sensitivity and scenario analyses. Infrastructure and policy measures are choices rather than parameters to estimate, and are explored in explicit scenarios. The design combines established types of analysis for agent-based models (Lee et al., 2015) in five steps (Figure 3):

1. **Calibrate plausible decision weights.** Candidate weight sets, sampled within the ranges from the literature, are scored against the observed adoption in 2022–2024, and the plausible ones are retained as an ensemble (Section 6).
2. **Rank the uncertain parameters.** Morris elementary effects establish which parameters most influence the 2050 heating mix.
3. **Quantify the key parameters.** The highest-ranked parameters are varied jointly in a Latin-hypercube sample.
4. **Run pathways and experiments.** The ensemble is carried through the scenarios at national scale, focusing on feedback loops, policy and infrastructure.
5. **Analyze segments.** Results are broken down by adopter category, ownership and dwelling, to show which groups drive each dynamic.

For the segment analysis, every owner-occupier household is assigned an adopter category following Rogers' diffusion framework (Rogers, 2003b), by percentile of an adoption index that combines its sustainability attitude, the mean attitude of its peers and the technical readiness of its dwelling, using the standard shares (innovators 2.5 %, early adopters 13.5 %, early and late majority 34 % each, laggards 16 %). Dwelling segments cross archetype, floor area, construction era and insulation label, and are additionally crossed with

ownership and heat demand.

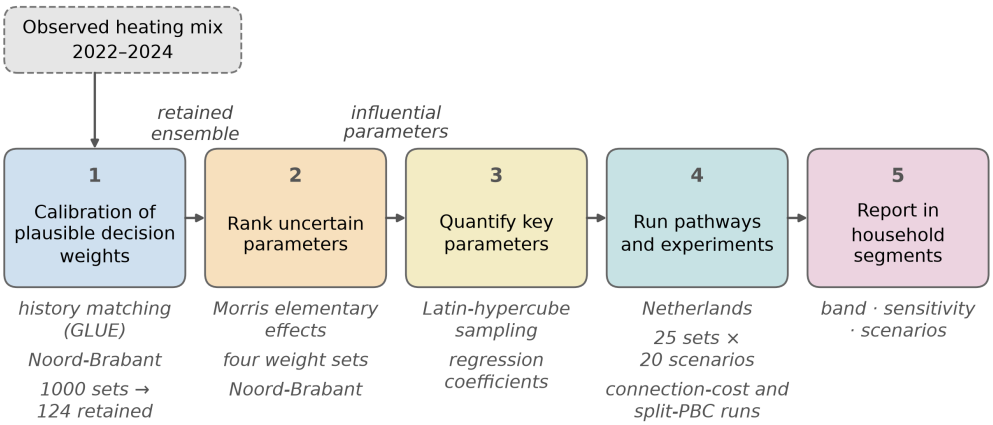


Figure 3: The five-step analysis design. Calibration and parameter screening are carried out on the calibration region (Noord-Brabant). The scenario pathways and mechanism experiments are run for the full Netherlands.

5.1. Verification and validation

The model is verified through unit tests on the decision functions, and a check that the initialized 2022 heating mix matches the observed mix. It is documented following the ODD protocol (Grimm et al., 2010) in Appendix A. Consistent with its purpose, the model is a tool to understand mechanisms and map plausible pathways rather than to predict them. Therefore, matching the observed 2022–2024 change filters out implausible parameterizations but does not validate the 2050 pathways.

6. Calibrating to plausible decision weights

We calibrate only the weights of the decision function and the share of the random term. They are central to adoption behavior and thereby to the feedback loops. Also, they determine the influence of the other uncertain assumptions. For example, when costs weigh little in the decision, uncertainty in costs matters little as well. Calibrating all parameters jointly would mainly reveal trade-offs between connected assumptions rather than create a coherent bounded set of weights. The other parameters are explored in the sensitivity analysis instead.

The only consistent data set on the adoption of heating methods in the Netherlands spans just three years (2022–2024). This short window cannot uniquely identify the decision weights, as many combinations reproduce recent adoption equally well. Rather than forcing a best fit and risking overfitting, we use history matching within the Generalized Likelihood Uncertainty Estimation (GLUE) framework (Beven, 2006). Each weight is sampled as a share within its stage of the decision function, over its whole feasible range, so that no prior judgement on the weights restricts the search. A Latin-hypercube sample of 1000 weight sets is drawn across this space and judged against the observations as described below.

Because each stage of the decision function is normalized by the sum of its weights, only their ratios matter, and the weight space is fully covered by four free parameters (attitude and subjective norm within intention, affordability within PBC, intention within behavior). The scale λ of the random term in the choice (Equation 5) is calibrated together with the weights as a fifth parameter. As the weights are normalized, λ sets how often a household chooses an option other than the one with the highest utility. Because the utility differences between options are small and the current system has the highest utility for most households, this largely determines how many households switch in a given year. It is sampled log-uniformly between 0.005 and 0.05. The retained sets form an ensemble that is carried into the scenario analysis, so that pathways are reported as a band across behaviorally plausible preferences rather than as a single fitted line.

Calibration is carried out on the province of Noord-Brabant over 2022–2024, the most recent window for which the Dutch statistics office reports a methodologically consistent per-neighborhood heating-mix series. The model is initialized from the observed 2022 mix, and each weight set is scored on the simulated 2023 and 2024 shares of natural gas boilers, hybrid and electric heat pumps and district heating. Following standard practice in history matching (Vernon et al., 2010), as applied to agent-based models by McCulloch et al. (2022), a set is retained unless one of these shares lies implausibly far from the observation, which is more than three times the combined uncertainty of the model's stochastic variation, the observation and the model's simplification, which amounts to about 1.7 percentage points (Appendix B.1).

6.1. The calibrated weights set

Of the 1000 sampled weight sets, 124 are retained (Figure 4), with a mean absolute deviation of 0.22–0.86 percentage points, against a median of 1.27

across all sampled sets. Above the cutoff, the deviation levels off at about 2.9 percentage points for around a third of the sets; in nearly all of these, the gas share stays too high because owner-occupiers switch too slowly.

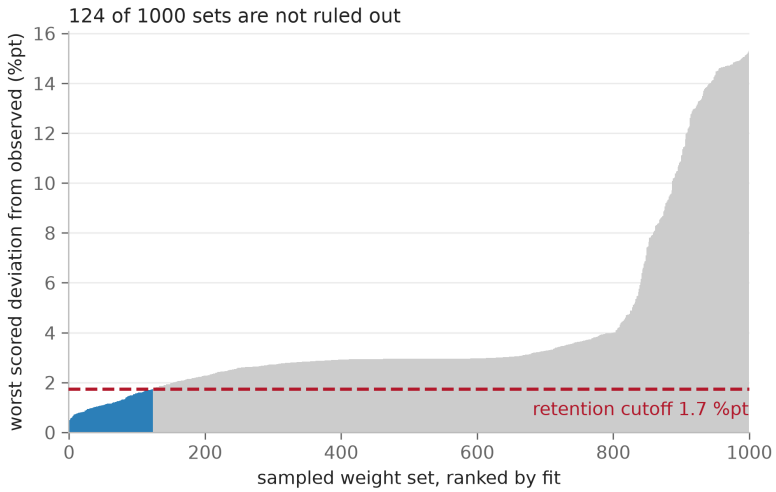


Figure 4: Worst scored deviation from the observation of all 1000 sampled weight sets, ranked, with the retained ensemble in color.

Taken one at a time, the decision weights are hardly narrowed by the observations, they constrain a combination of the weights instead (Figure 5). The attitude and affordability shares are strongly opposed, and require a sum which for the formula $w_{\text{cost}} + 2.0\, w_{\text{att}}$, lies between 0.62 and 0.91 (Figure 5. This follows from the decision function. In this early phase, effort and the subjective norm favor the incumbent gas boiler, while attitude and, to a lesser extent, cost favor heat pumps. As the effective weights sum to one, weight given to cost or attitude is taken from effort and peers, so their combined weight determines how many households switch. Too low a combination keeps the gas share too high; too high a combination leads to too much adoption. Within these bounds, the individual weights can take nearly their full range (Table 3).

The observation window thus fixes how strongly owner-occupiers are pushed towards a new heating method in total, not where that push comes from. Cost and attitude substitute for each other, one unit of attitude weight does the same as about two units of cost weight. The subjective norm hardly contributes in this window, as the peer signal is weak while heat pumps are still rare, which is why it is only bounded from above. The ensemble therefore

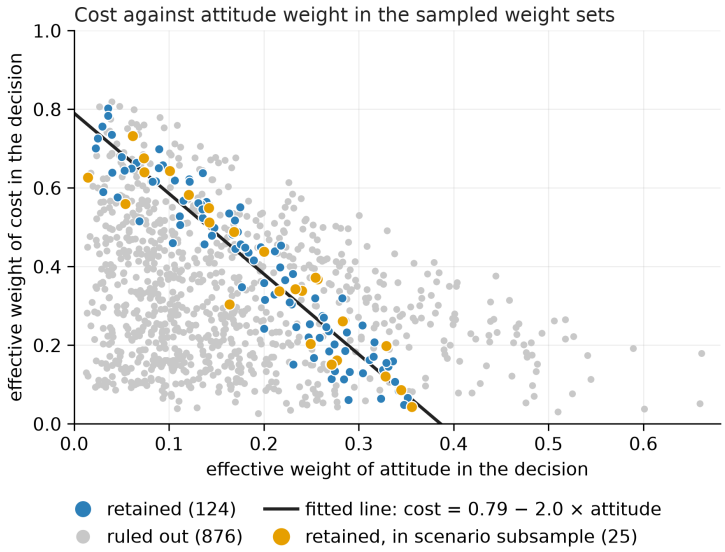


Figure 5: Effective weight of cost against effective weight of attitude in the final decision, for all 1000 sampled weight sets. The retained sets in blue lie along the cost-attitude line. The observations constrain the trade-off between the two weights rather than either weight on its own. The line is a least-squares fit through the retained sets. The 25 sets used in the scenario analysis are displayed in orange.

Table 3: Range of each decision weight in the retained ensemble against the range it was sampled from, as normalized shares within their stage of the decision function, and the same for the combination that the observations constrain along the cost–attitude line. The last column gives the width of the retained range as a share of the sampled range.

Weight	**Sampled**	**Retained**	**Retained width**
Attitude (of intention)	0.10–0.77	0.10–0.72	92 %
Subjective norm (of intention)	0.10–0.76	0.10–0.64	82 %
Affordability (of PBC)	0.10–0.90	0.12–0.90	98 %
Intention (of behavior)	0.10–0.90	0.10–0.88	98 %
Noise scale λ (log scale)	0.005–0.050	0.005–0.049	97 %
$w_{\text{cost}} + 2.0\, w_{\text{att}}$ (effective)			21 %

spans behaviorally distinct households, from cost-driven to attitude-driven, that are expected to respond differently to price and subsidy changes.

The observations do not pin the noise scale down either. In the retained sets, between about 1 and 16 % (median 5.5 %) of the owner-occupier decisions in 2023–2024 fall on an option other than the one with the highest modeled utility. Simulating all 124 retained sets for Noord-Brabant until 2050 shows that the noise scale mainly affects hybrid heat pumps. Where more random choice leads to more hybrids, which are often the close second option behind a gas boiler or an electric heat pump.

6.2. Selecting a subsample

Running all 124 retained sets through every scenario at national scale is not feasible, so the scenario analysis uses a subsample of 25 sets, chosen so that the median and the 5–95 % range of the baseline pathway in Noord-Brabant match those of all retained sets (Appendix B.1). The resulting bands describe the spread across plausible weight sets, not probability intervals. The sensitivity analysis draws its sets from the same subsample. Because the influence of a parameter can depend on the weights, the Morris screening is repeated at four sets that span the ensemble: the best-fitting set, the most cost-driven and the most attitude-driven set, and the set with the strongest subjective norm. The Latin-hypercube sample uses the best-fitting set.

7. Results

The Netherlands is simulated over 2024–2050 for each of the 25 weight sets across twenty scenarios. The sensitivity analysis is done for the province of Noord-Brabant. Pathways are reported as ensemble bands and scenario effects as the median across the ensemble. After the baseline, the results follow the dynamics of Table 1: the sensitivity analysis and the feedback loops (Sections 7.1 and 7.2), the decisions of different groups of actors (Section 7.3), policy and infrastructure (Section 7.4), and an alternative decision structure (Section 7.5).

The central result is that the decline of natural gas is robust. It occurs in every scenario and for every weight set. How far it declines, and which technologies replace it, depend on behavior, policy and infrastructure, and no scenario phases out natural gas fully by 2050. In the baseline (Figure 6), natural gas falls from 86 % of dwellings in 2024 to a median of 15 % in 2050 (5–95 % range 7–27 %). Electric heat pumps become dominant (median 57 %,

47–70 %), while hybrid heat pumps (16 %, 7–26 %) and district heating (11 %, 7–19 %) take smaller shares. The spread comes mainly from the trade-off between the cost and attitude weights and from the noise scale (Section 6.1).

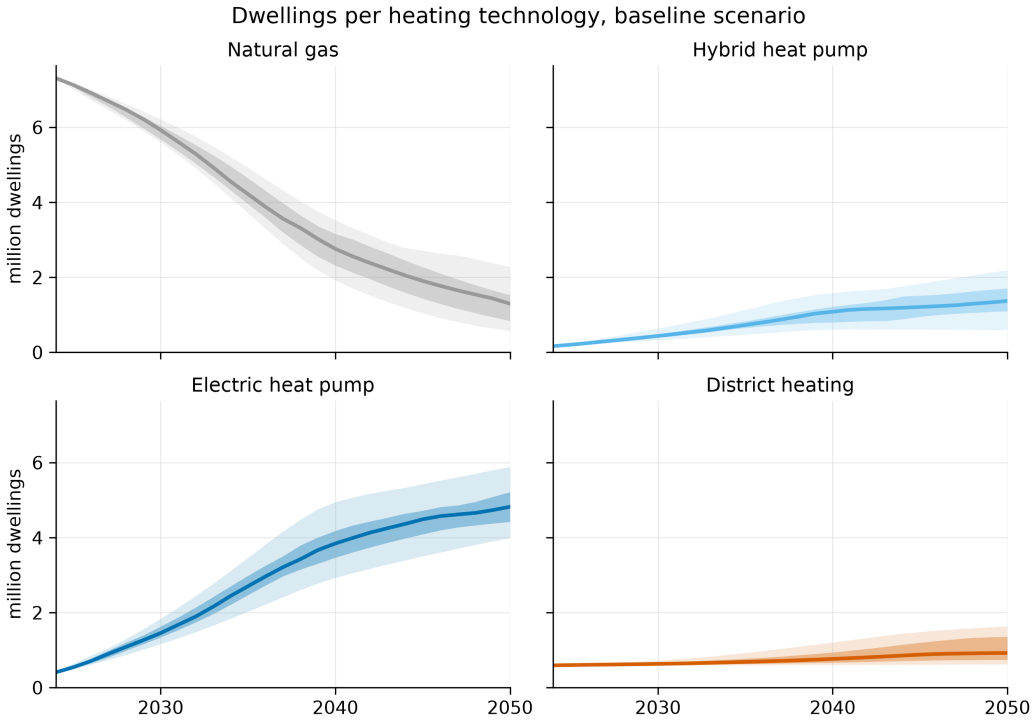


Figure 6: Number of dwellings per heating technology in the baseline scenario. The line is the median across the 25 weight sets, the dark band the 25–75 % range and the light band the 5–95 % range. Natural gas includes individual and collective gas boilers.

7.1. Sensitivity analysis

7.1.1. Identifying relevant parameters using Morris elementary effects

We first rank the uncertain parameters with the Morris elementary-effects method (Morris, 1991; Campolongo et al., 2007), repeated at the four screening weight sets (Section 6.2). The design, the parameters and their ranges are given in Appendix C.1. Figure 7 ranks the parameters by their effect on the 2050 mix, colored by the part of the decision function they enter. Because the effect is measured across the range of each parameter, the ranking depends on the ranges. Those of the cost parameters are taken from the

literature, whereas those of the social and attitude parameters are modelling choices, for lack of data.

Social learning has by far the largest effect, mainly on the gas share. Stronger perceived norms favor the technology that currently dominates, and thus reinforce the natural gas based regime. The attitude distribution and the perceived sustainability of the hybrid heat pump follow. This perceived sustainability decides between the two heat-pump types. If hybrids are seen as a credible intermediate, they are widely adopted. If they are not perceived as a relatively sustainable alternative, the electric heat pump is preferred. The learning rate, the heat-pump capital cost, the opportunity trigger, the network size and the energy prices form a second group, while the salience parameters matter much less. District heating is insensitive to all parameters tested, as without strong policy measures there is little adoption in any case.

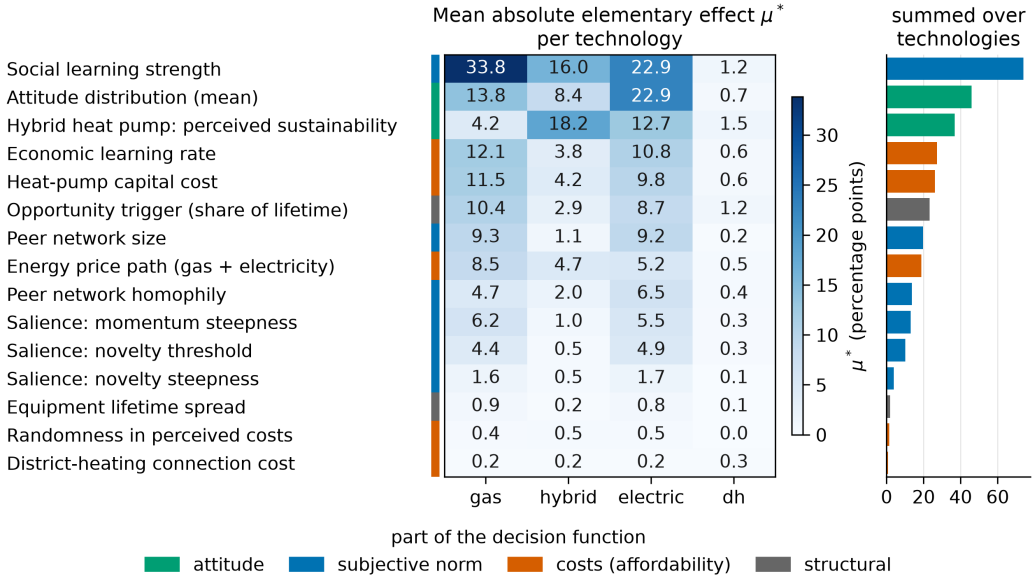


Figure 7: Sensitivity of the 2050 heating mix in Noord-Brabant to the screened parameters, ranked by their total effect. Values are the mean absolute elementary effect μ^* of the Morris method, in percentage points. Which is the change in a technology's 2050 share when a parameter is moved across its range with all others held fixed, averaged in absolute value. The color strip shows the part of the decision function each parameter enters, structural parameters affect when a decision is made rather than how it is made.

7.1.2. Joint variations of key parameters

The seven highest-ranked parameters are then varied jointly in a Latin-hypercube sample (McKay et al., 1979) of 80 combinations, eight iterations each, at the best-fitting weight set. Their direction, magnitude and interactions are assessed by regression (Saltelli et al., 2008). Together they spread the 2050 share of hybrid heat pumps over 8–48 % and that of electric heat pumps over 35–74 % (5–95 % range, Figure 8). The split between the two heat pumps is set mainly by the perceived sustainability of the hybrid, which across its range moves about 35 percentage points to hybrid heat pumps and by the attitude distribution. Greener attitudes add about 20 points to the electric heat pump. The gas share, 2–22 %, depends on three parameters of similar size, each moving it by 11–13 points across its range. Stronger social learning and a later opportunity trigger keep more gas boilers in place, while faster learning reduces them. Greener attitudes and cheaper heat pumps reduce it by a further 6–8 points. District heating stays between 8 and 12 %, and the network size has little effect at this weight set. The parameters act largely independently, as a linear model explains 74–90 % of the variation in each share. Interactions matter mainly for gas, raising this to 95 %, chiefly between the learning rate and the heat-pump capital cost and between social learning and the opportunity trigger.

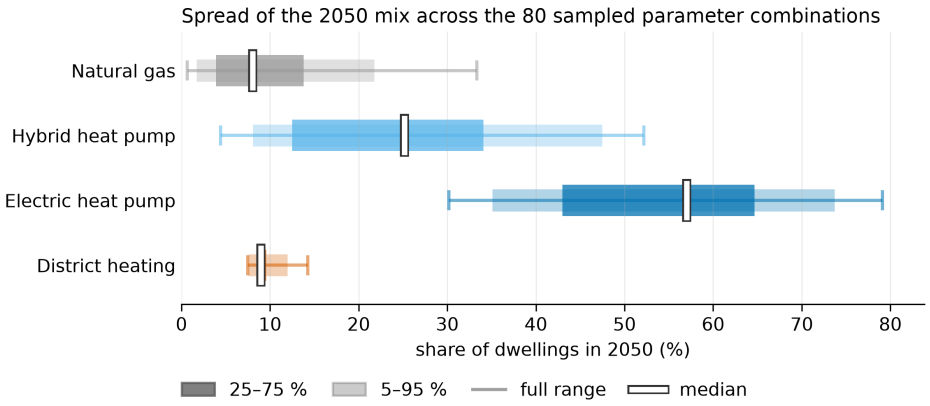


Figure 8: Spread of the 2050 heating mix in Noord-Brabant across 80 combinations of the seven highest-ranked parameters (Latin-hypercube sample, eight iterations each), at the best-fitting weight set. The graph shows the full range, 5–95 % and 25–75 % ranges, and median.

Taken together, the screening and the joint variation separate robust

from uncertain outcomes. In every combination, natural gas declines without disappearing and district heating stays small without policy. How far gas declines depends mainly on social learning, economic learning and the timing of replacement. Across the ensemble, social learning is the strongest of these, but its effect depends on the weight households give to the subjective norm. It dominates where that weight is high and vanishes where decisions are driven by cost, whereas the effect of economic learning is similar across weight sets. The split between hybrid and electric heat pumps depends mainly on how sustainable the hybrid is perceived to be and on households' attitudes. The parameters that matter most are thus the social and attitudinal ones whose ranges rest on modelling choices rather than data, which makes them a priority for empirical research. Their effect on the 2050 mix is of the same order as that of the uncertainty in the decision weights.

7.2. Feedback loops

The sensitivity analysis shows that the parameters driving the economic and social learning loops are key to the results. To understand when and by whom these dynamics are triggered, we vary the strength of each loop in the scenario analysis (Figure 9).

Social learning is the strongest lever, in line with the sensitivity analysis. It favors the status quo, with strong social learning, 26 % of dwellings still heat with natural gas in 2050, against 10 % with weak social learning and 15 % in the baseline. The 2050 share of electric heat pumps differs by only half a percentage point, so the difference in gas is taken up mainly by hybrid heat pumps and partly by district heating. Most of the effect, however, lies in the pathway rather than in the 2050 shares (Figure 10). With strong social learning, electric heat pumps fall up to 8 percentage points behind the baseline around 2036, but this gap has nearly closed by 2050, whereas the gas share keeps diverging until the early 2040s and remains above the baseline in 2050. Peer influence first favors the dominant technology, but as electric heat pumps spread, the norm shifts towards them.

The economic learning loop has no threshold of its own, as every installed unit slightly reduces the costs, and because households differ widely in energy demand and insulation costs, the resulting adoption curves are smooth. Strong economic learning reduces the 2050 share of natural gas to 8 %, against 19 % with weak learning. It interacts with other cost drivers, with low gas and high electricity prices, 21 % of dwellings still use natural gas in 2050, against 7 % with high gas and low electricity prices. In the scenario most

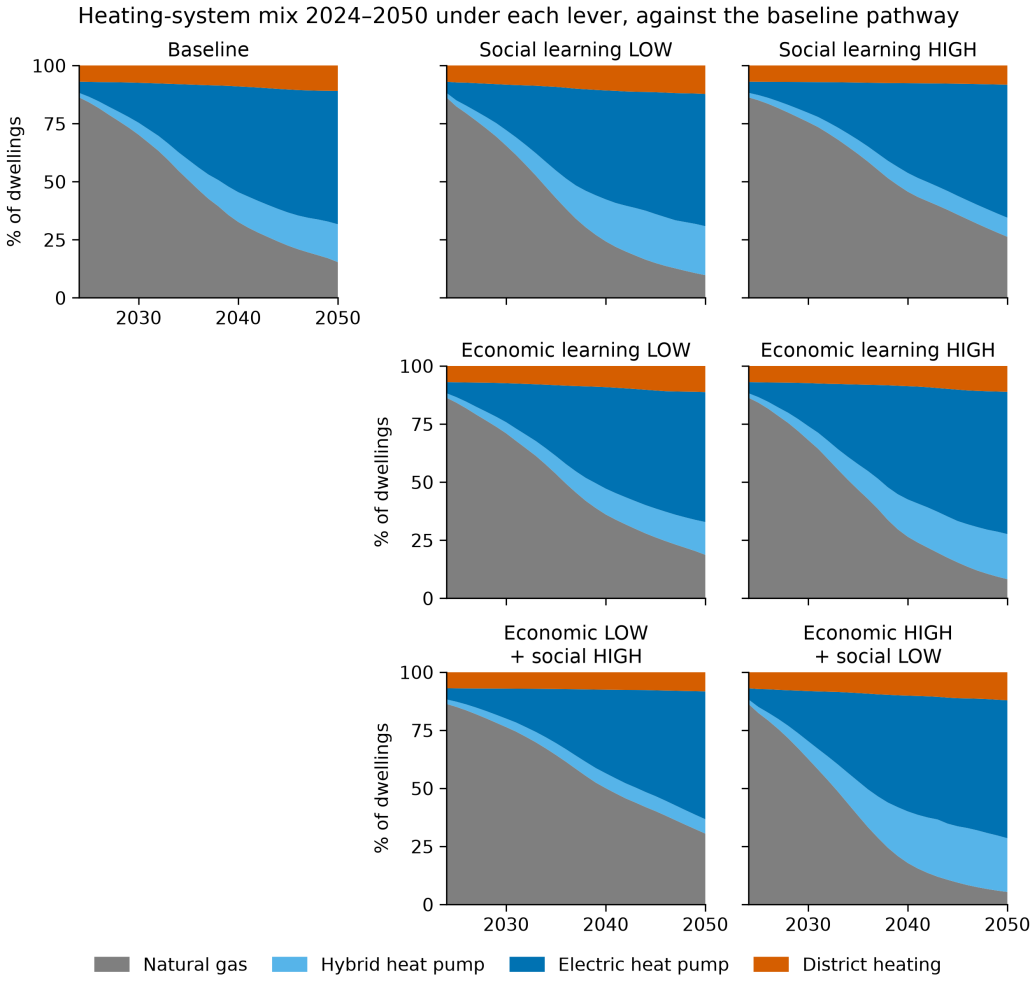


Figure 9: Heating mix 2024–2050 in the feedback-loop scenarios (median across the 25 weight sets). Scenario's include weak and strong social and economic learning and the two loops combined in opposite directions. Stronger social learning favors the status quo, and stronger economic learning reduces natural gas.

favorable to natural gas, with strong social learning, weak economic learning and cheap gas, 41 % of dwellings still heat with gas in 2050, against 2 % in the scenario most favorable to heat pumps.

The two loops do not reinforce each other, their effects roughly add up. With weak economic and strong social learning, 31 % of dwellings still use natural gas in 2050. In the opposite combination, 5 %. Even in the most favorable scenario, natural gas does not disappear. None of the feedback

loops has a strong effect on district heating, whose 2050 share stays between 8 and 12 %, indicating that heat networks will not expand without policy action.

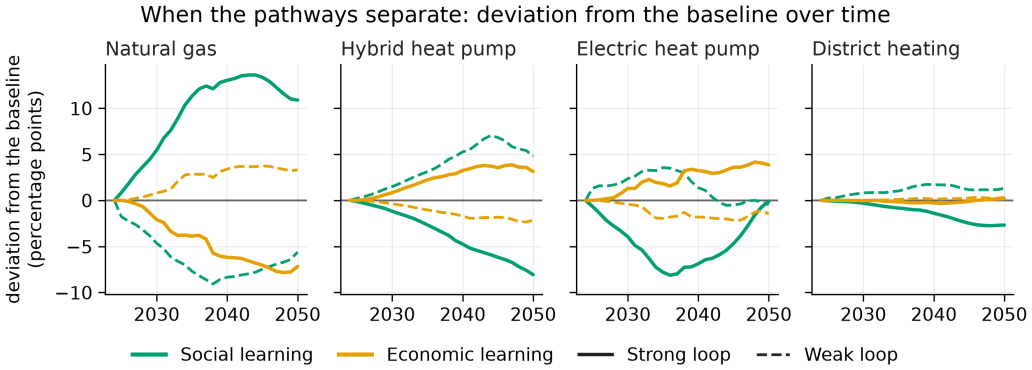


Figure 10: Deviation of the heating mix from the baseline in the feedback-loop scenarios, in percentage points (median across the 25 weight sets).

7.3. Adoption per adopter and dwelling category

To see which groups drive these dynamics, we analyze the 2050 heating mix per segment in the baseline scenario (Figure 11). The figures in this section show the best-fitting weight set, how the outcome splits between ownership types differs between weight sets.

Ownership determines which decision rule applies (Figure 11a). Owner-occupiers, who weigh attitudes and norms as well as costs, end with the largest share of hybrid heat pumps (34 %) and almost no natural gas (3 %). Housing associations and private landlords, who decide on cost alone, electrify at least as much (60 and 58 % electric heat pumps, against 51 % for owner-occupiers), but where the electric heat pump is not the cheapest option they keep their gas boilers, 12 and 10 % of their dwellings still use natural gas in 2050. This shows how these actors would act without the sustainability requirements, rules and incentives they face in practice. District heating reaches 12–13 % of owner-occupied and social-housing dwellings, but only 8 % of privately rented dwellings and none of the dwellings in owner associations.

Within the owner-occupied stock, the adopter category separates the outcome far more than any dwelling characteristic (Figure 11b). Innovators, early adopters and the early majority choose the electric heat pump almost exclusively (83–92 %). The late majority turns mainly to hybrid heat pumps

(60 %), and the laggards almost entirely so (67 % hybrid), while 17 % of the laggards still heat with natural gas in 2050. The lagging half of the owner-occupiers thus act in the transition, but mainly by means of hybrid heat pumps.

Heat demand and dwelling type mainly shift the choice between hybrid heat pumps and district heating (Figure 11c–d). Owner-occupied dwellings with a low heat demand, often apartments and high-rise, connect more often to district heating, whereas dwellings with a high demand take more hybrids, while the share of electric heat pumps hardly changes. The insulation label separates the outcome more strongly (Figure 11e). Two-thirds of dwellings with label A or B in 2024 have an electric heat pump in 2050, against one-third of those with label E to G.

The adoption process is not finished in 2050 (Figure 12). Electric heat pumps spread rapidly among the innovators and early adopters and level off around 2040 at about 90 %. The early majority follows a few years later. In the late majority, electric heat pumps reach about a quarter of the dwellings around 2045 and then stagnate, while hybrid adoption in the late majority and the laggards is still increasing in 2050, and district heating grows from about 4 to 15 %.

Comparing the cost-optimal option, with the option of highest utility, and the actually adopted option, shows a familiar transition dynamic (Figure 13). Electric and hybrid heat pumps are already the cost-optimal option for three quarters of installations in 2030, yet natural gas has the highest utility in more than half of them. Through attitude, social norm and effort, owner-occupiers stay with the incumbent system even when an alternative is cheaper. The gap closes as attitudes and norms towards heat pumps grow. Because of the random term, a dwelling whose highest-utility option leads only narrowly may choose the runner-up. Natural gas is involved in most of these close decisions, especially early-on in the transition.

7.4. Policy and infrastructure

Next, we test whether policy measures and infrastructure can trigger or stop these dynamics. We start this section with exploring the impact of a ban on electric heat pumps in neighborhoods with grid congestion, which arises from the combined load of heat pumps and electric vehicles. This does not slow the transition away from gas and does not change the share of district heating (Figure 14). Instead, it substitutes the share of electric heat pumps for hybrids. Households that cannot install an electric heat pump turn to

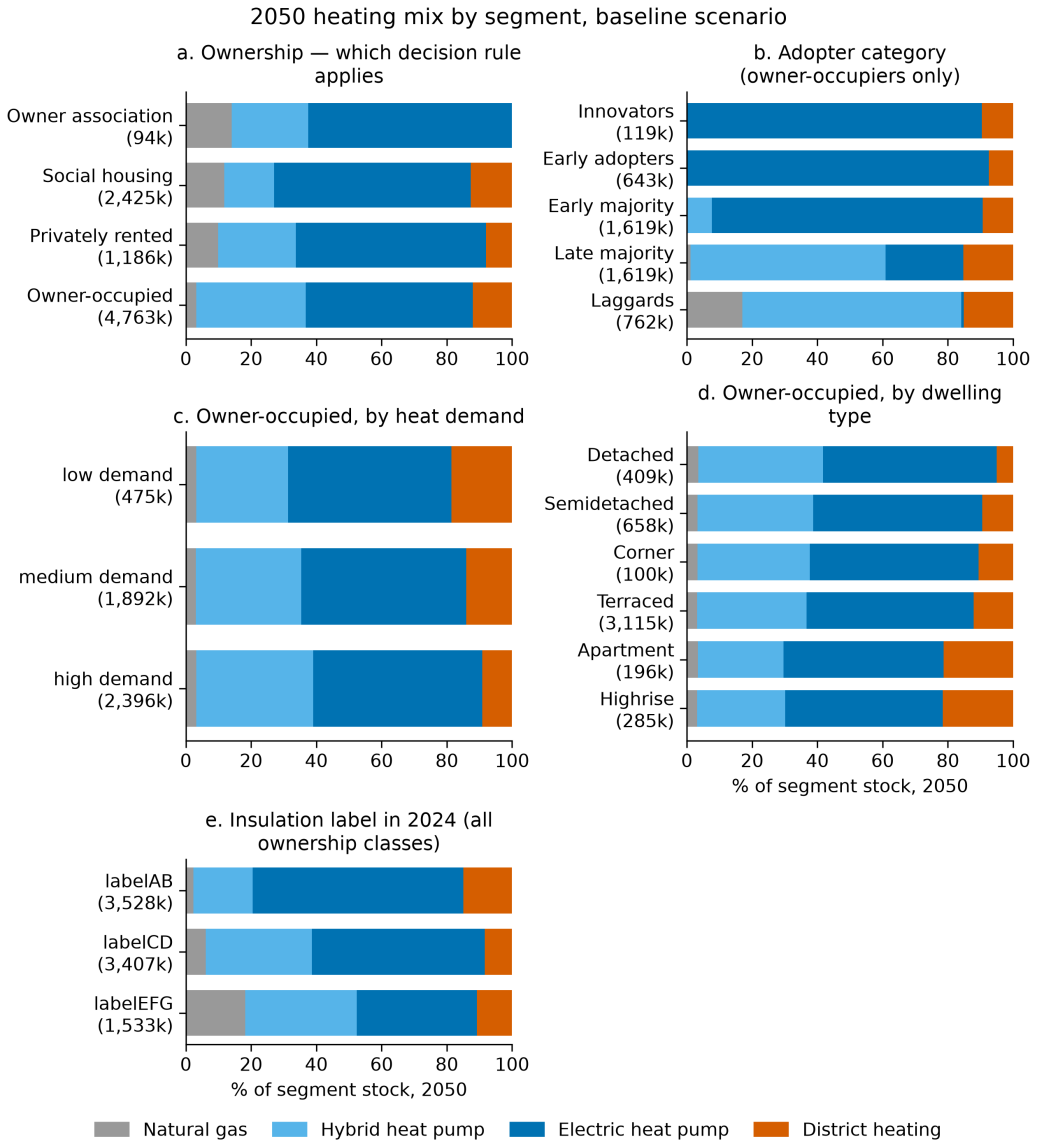


Figure 11: 2050 heating mix by segment in the baseline scenario, best-fitting weight set. (a) ownership, (b) adopter category of owner-occupiers, (c–d) heat demand and dwelling type of owner-occupied dwellings, (e) insulation label in 2024.

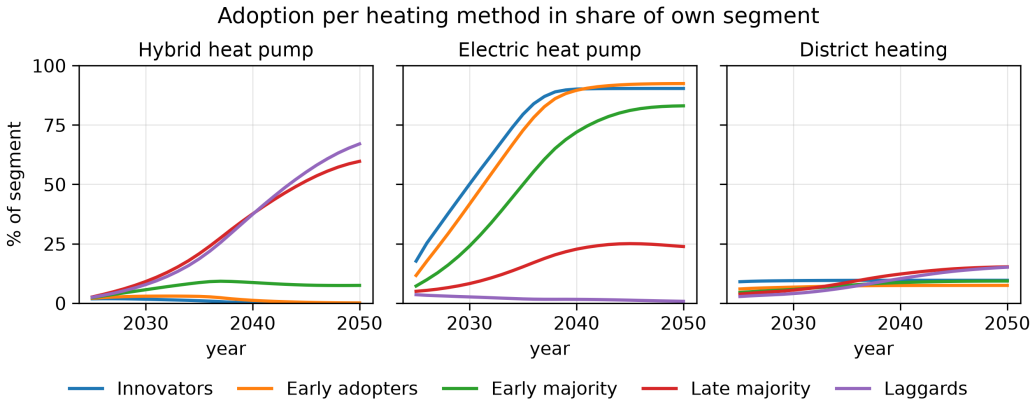


Figure 12: Adoption by adopter category, per technology, as a share of each segment's own stock, in the baseline scenario for the best-fitting weight set.

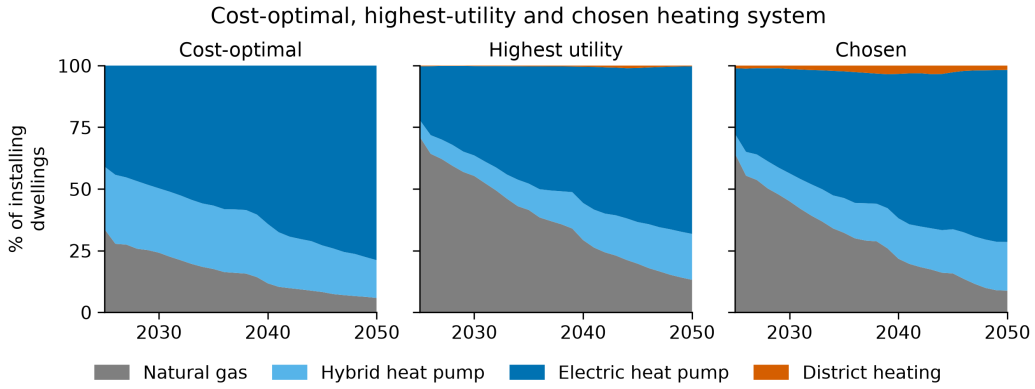


Figure 13: Cost-optimal (lowest EAC), highest-utility (before the random term) and adopted heating system in the baseline scenario, as a share of the dwellings that install a heating system in that year.

its closest alternative, not only in cost but also in attitude and norm. In the model, congestion spreads in the early 2030s, reaching about half of all neighborhoods by 2035. This is later than in reality, where it is an immediate issue, but the dynamic makes the hybrid heat pump a potential transition technology where the grid requires it.

District heating hardly expands on cost alone. We therefore test two further scenarios. First, heat networks built in the areas selected by municipalities, regardless of household demand. Second, these same networks combined with a connection obligation building on new regulations towards

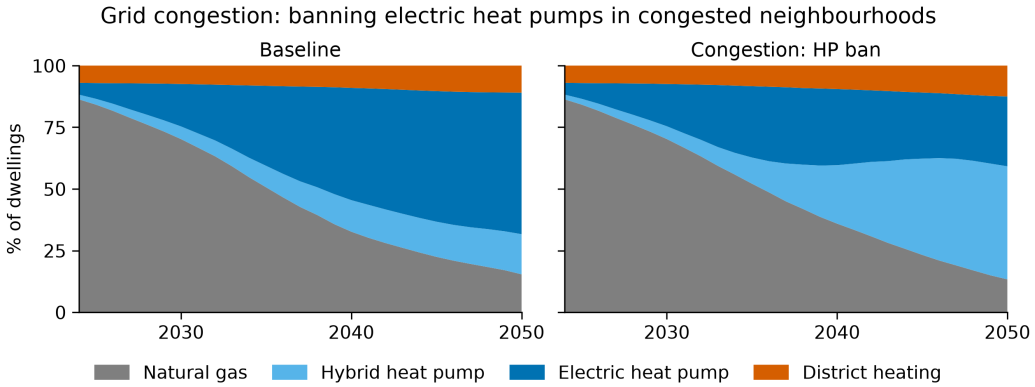


Figure 14: Heating mix 2024–2050 with and without a ban on electric heat pumps in congested neighborhoods (median across the 25 weight sets).

district heating (Figure 15). When connecting remains voluntary, few households connect and district heating ends at about the baseline share. Only with an obligation does district heating expand, to about 40 % of dwellings by 2050, mainly at the expense of the electric heat pump. Putting the infrastructure in place is therefore not enough, an obligation, stronger incentives to connect, or much cheaper district heating tariffs are required.

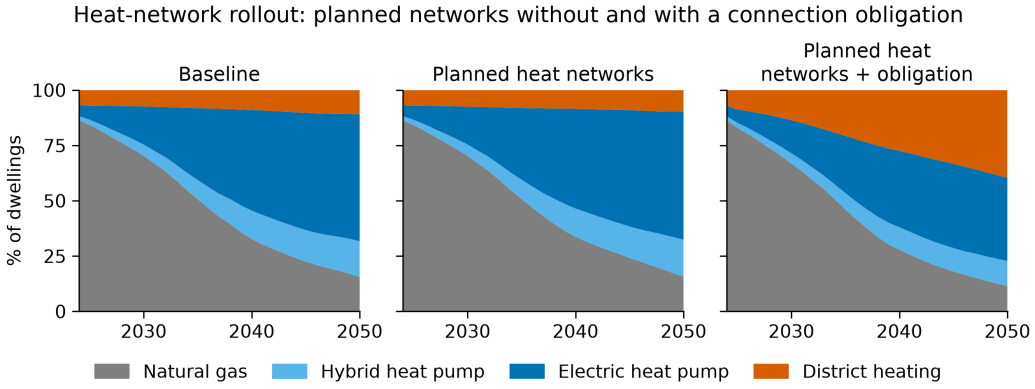


Figure 15: Heating mix 2024–2050 in the baseline, with the heat networks planned by municipalities, and with the same networks combined with a connection obligation (median across the 25 weight sets).

Finally, we examine who pays for the energy networks. Under current Dutch tariffs, the fixed costs of a network are paid by whoever is connected to it. Households on district heating pay for their heat network and also for

their electricity and sometimes gas connection, whereas households with an electric heat pump pay only for the electricity grid. This allocation is part of an ongoing political debate (Verstraten and Huygen, 2025; Spies et al.), and electricity tariffs will become partly consumption-based from 2029 (ACM, 2026; Common Futures, 2026). We explore the far end of this debate with a scenario in which all gas and district-heating connection costs are removed, which is equivalent to spreading them evenly across all households, while equipment investments and energy prices are unchanged. The removed costs are the annual gas connection charge (transport and fixed charge), the annual district-heating charges (fixed charge, metering and delivery-set rental) and the one-off district-heating connection and disconnection fees The result is a substantially different heating system (Figure 16). By 2050 the share of electric heat pumps falls from 57 to 17 %, while that of hybrid heat pumps rises from 16 to 33 % and that of district heating from 11 to 40 %. Without the gas charge, the hybrid heat pump, with its lower investment, becomes cheaper than the electric heat pump. Also, without its connection charges, district heating becomes the cheapest option for many dwellings within reach of a network. The shift is carried mainly by housing associations and landlords, who decide on cost. Among owner-occupiers, attitudes and peers keep about half of the decisions in which the hybrid has become cheapest with the electric heat pump.

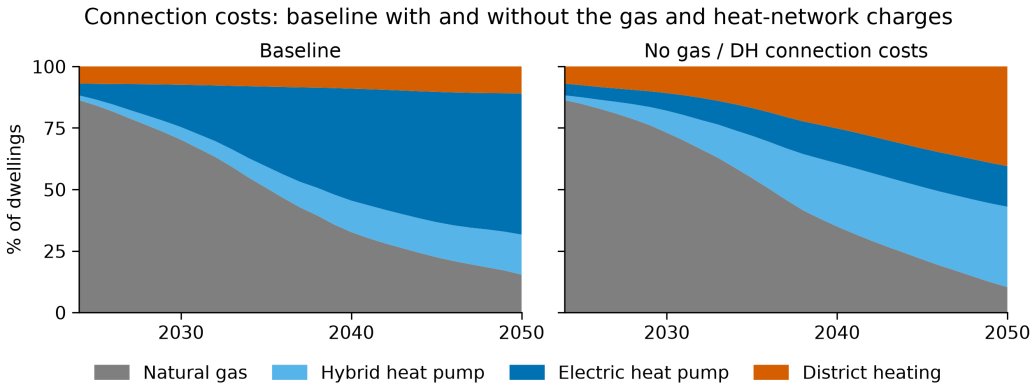


Figure 16: Heating mix 2024–2050 in the baseline (left) and without gas and district-heating connection costs (right), median across the 25 weight sets.

7.5. Split investment and running costs in decision-making

Lastly, we explore structural uncertainty in the decision function itself. An often mentioned barrier is the high upfront investment in heat pumps (Jager et al., 2025; de Jong et al., 2024; Gaur et al., 2021; Ebrahimigharehbaghi et al., 2019), especially when a renovation is required (Lamb and Elmes, 2024). Even when their equivalent annual costs are lower, households may be unable or unwilling to pay the investment. We test this by splitting the cost term in the PBC (Equation 2) into the annual running costs and the upfront investment costs. Each with its own weight, and recalibrating the weights on the same observations. The split structure fits the observations as well as the original one, and the cost–attitude trade-off reappears, now carried by the running-cost weight. The decline of natural gas is nearly unchanged, but district heating rises about 7 %, at the expense of electric heat pumps (Figure 17). District heating, with its subsidized connection fee, has low upfront but relatively high running costs, so it gains wherever households weigh the investment separately. While electric heat pumps have large investment costs, especially for those dwellings who require renovations.

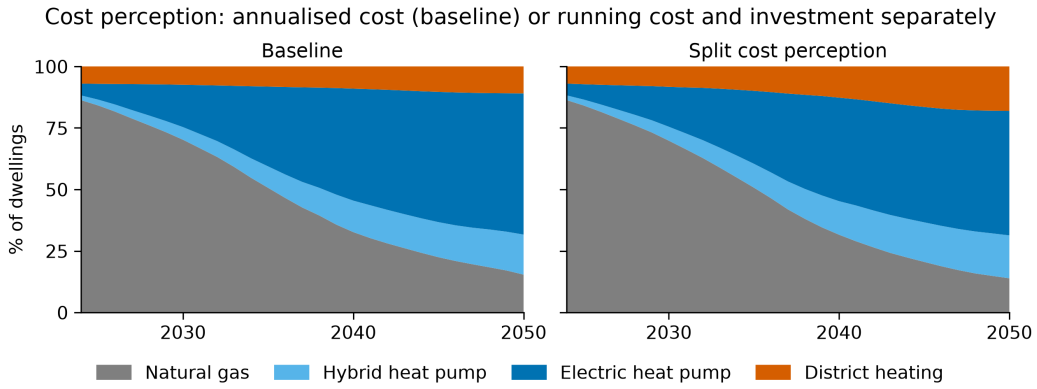


Figure 17: Baseline heating mix with owner-occupiers' cost perception as one annualized cost (left) or as annual running cost and net upfront investment scored separately (right). Median across each recalibrated decision-making weight sets.

8. Discussion

This paper shows how bottom-up modelling, grounded in individual decisions in a heterogeneous population, can be used to explore the transition

pathways of an entire sector. It produces non-linear adoption that can be studied per population group, ownership type and dwelling segment, and it allows a broad exploration of the solution space. Such models should nevertheless be used with caution. Actual decision-making is more elaborate and context-specific, and people use different decision strategies (Michelsen and Madlener, 2013) with large heterogeneity in preferences (Meles et al., 2022), and a decision as consequential as a heating system usually evolves over time through social discourse and information seeking. The uncertainty of such processes has led most energy-transition pathway studies to omit them. We argue that it is better to include them and explore their influence across a wide range of plausible parameterizations, which leads to the following points.

Social reinforcement favors the incumbent. Strengthening social learning slows the transition rather than accelerating it. This runs against the common framing of social influence as a lever for accelerating low-carbon adoption, in which peer effects and social norms are mechanisms to be amplified (Wiedermann et al., 2020; Lenton et al., 2022). The contradiction is caused by the direction of the mechanism. Social influence reflects what peers currently do, which at the start of the transition is the incumbent system. Strong social learning therefore delays the spread of electric heat pumps. Once they have gained ground, the norm turns in their favor and the delay is largely made up by 2050. Although norms form within local, small-world networks, the net effect is still driven by the majority. This would only differ if social influence were asymmetric, with adopters of a new technology weighing more heavily as peers than users of the incumbent, which can be explored with the same model in future work.

The two loops work differently. Economic learning acts in the same direction from the start: every installed heat pump lowers the cost of the next, and adoption follows smoothly, without a threshold. Social learning follows the current majority and therefore changes direction during the transition. Combined, the two loops add up rather than amplify each other, while the parameters within a loop do interact.

Rules and policies shape the route. Natural gas declines in every scenario and weight set but disappears in none, and under current rules the electric heat pump becomes the dominant alternative across all plausible decision weights. Rules and infrastructure change the route where hybrids are often a good second-best alternative between natural gas and all-electric, and district heating requires strong policy support.

Social and psychological parameters remain hard to pin down. The parameters that most widen the range of pathways describe social influence, perceived sustainability, and attitudes, which are also the parameters with the least empirical support. The same holds for the decision structure, how households weigh upfront investment against running costs decides between electric heat pumps and district heating, while these differences are very context and individual dependent. Better data on how households perceive and discuss heating options would narrow the uncertainty more than better cost data, in line with calls for the empirical validation of behavioral insights in agent-based models (Rietkerk et al., 2025). However, inherent to the complexity of decision-making, uncertainty will remain. Models can be used to explore different pathways while acknowledging this uncertainty.

8.1. Limitations

The calibration window is short and atypical. Two years of observed change bound the decision weights only partly, which is why we report an ensemble rather than a fitted parameterization. The window coincides with the gas-price crisis and strong subsidies, while the model holds energy prices at their 2025 level over this period.

Matching the recent past does not validate the projections. The calibration filters out implausible parameterizations but does not establish that the 2050 pathways are correct. These are interpreted for the mechanisms they reveal, not as forecasts.

Structural uncertainty is explored only partly. We test one alternative form of the cost perception (Section 7.5), but not alternatives to the theory of planned behavior as a whole or to the description of social visibility, and the sensitivity results are conditional on parameter ranges that are partly modelling choices.

Some mechanisms are represented schematically. Peer networks include owner-occupiers only, district heating has no economic learning and a regulated tariff, and landlords and housing associations are simplified cost-minimizing agents. The segment analysis reports a single weight set, and the sensitivity analysis is performed for one province. Additional detail in all these cases would be possible in future studies, this way, enabling scholars to dive deeper into the adoption patterns of the empirical dwelling stock finding more transition dynamics that could be used to guide policy decisions.

9. Conclusion

We set out to answer two questions: which features of the Dutch residential heat transition are robust across plausible behavior and which depend on behavior, policy and infrastructure, and what an empirically grounded bottom-up model can contribute to transition planning when data are limited and future decision-making is inherently uncertain.

On the first question, the decline of natural gas is robust, but its replacement is not. Gas declines in every scenario and weight set but disappears in none, and under current rules the electric heat pump becomes the main alternative across all plausible decision weights. How far gas declines and what replaces it depend on behavior, policy and infrastructure. Strong social learning keeps gas in place for longer, a congestion ban redirects adoption to hybrid heat pumps, heat networks are used at scale only under a connection obligation, and the allocation of network costs can shift the mix from electric heat pumps to hybrids and district heating.

On the second question, such a model is most useful not for forecasting a single pathway, but for separating robust from contingent developments, locating the barriers to desired outcomes and showing what it takes to align household decisions with policy goals. Four features of the approach make this possible. First, retaining an ensemble of plausible decision weights instead of a single calibrated parameterization makes the limits of the data explicit. Two years of observed adoption admit 124 of 1000 sampled weight sets, which agree that gas declines but differ in how far and in the split between electric and hybrid heat pumps. Carrying them through all experiments separates no-regret developments from outcomes that depend on assumptions. Second, representing decisions and feedback loops explicitly reveals mechanisms that aggregate models miss. Social learning follows the current majority and therefore delays electrification early in the transition, whereas economic learning favors heat pumps from the start. Third, the bottom-up structure shows whom policy needs to reach. The leading half of owner-occupiers electrifies almost completely, the late majority and laggards turn mainly to hybrid heat pumps, and landlords and housing associations keep gas where the electric heat pump is not the cheapest option. Fourth, the sensitivity analysis shows where better data would help most. The parameters that most widen the range of pathways, such as social influence, perceived sustainability and attitudes, are the least supported empirically.

For policy, timing measures matters as much as instrument choice. Cost-

side measures move households away from gas at every stage of the transition, social instruments pay off once sustainable heating has a foothold, and new infrastructure needs obligations or incentives to be used. Further work should collect data on these social and psychological parameters, test alternative decision-model forms, including heterogeneous and evolving frameworks, and study spill-overs between ownership groups and the decisions of landlords and housing associations.

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Appendix A. Model documentation

This appendix documents the model following the ODD (Overview, Design concepts, Details) protocol (Grimm et al., 2010). Cost, learning and scenario details are given in Appendices Appendix A.2–Appendix A.4. Here we summarize the structure and refer back to the main text for the decision equations.

Appendix A.1. ODD protocol

Appendix A.1.1. Overview

Purpose and patterns. The purpose of the model is to explore which features of the 2024–2050 heating-technology pathways of the Dutch residential heat transition are robust across plausible household behavior, and which depend on behavior, policy and infrastructure. To this end, it simulates the heating decisions of owner-occupiers, landlords and housing associations for every dwelling, together with economic and social learning, grid congestion and the expansion of heat networks. The model is not meant to predict a single pathway. Because the available adoption data cannot pin down the decision weights, it is run with an ensemble of plausible weight sets. Its pathways are used to identify developments that hold across this ensemble, the barriers to desired outcomes, and the policies and infrastructure that change the route of the transition.

The model is evaluated against three patterns. First, low-carbon heating technologies should diffuse non-linearly, driven by cost decline and social influence. Second, adoption timing and technology choice should differ across household types, ownership and dwelling characteristics. Third, the simulated change in the provincial heating mix over 2022–2024 should match the observed change in direction and magnitude. This pattern is used to select plausible weight sets during calibration (Section 6).

Entities, state variables and scales. The model contains six entity types: dwellings, owner-occupiers, social housing associations, landlords, a district heating company and an electricity grid operator. On top of this there are neighborhoods as the spatial unit at which collective (grid and district-heating) decisions are made, and streets or housing blocks as the unit on which housing association and block heating decisions are made. Table A.4 lists the principal state variables per entity. Global state comprises, per heating technology, the learning-updated investment cost, cumulative installed

units, energy price and salience factor. The model advances in annual time steps over 2024–2050 (and over 2022–2024 for calibration). The spatial extent is a single province (the calibration and analysis scope, e.g. Noord-Brabant) or the full Netherlands (∼8 million dwellings). Dwellings are located to neighborhoods but not to coordinates.

Table A.4: Principal state variables by entity. Variables marked ∗ change during a run; others are set at initialization.

Entity	**State variables**
Dwelling	construction year, type, floor area, ownership, energy label∗, heat demand∗, in-house heat-distribution system∗, current heating method∗, system age∗, drawn end-of-life age.
Owner-occupier	sustainability attitude, peer network, per-method attitude, subjective norm∗ and perceived behavioral control∗, resulting intention∗ and utility∗.
Social housing assoc. / landlord	set of owned dwellings, adoption strategy (cost- or policy-based, for associations).
District heating company	expansion strategy, per-neighborhood grid state∗, construction lead time.
Grid operator	per-neighborhood peak demand∗ and congestion state∗, reinforcement rate.
Neighborhood	dwelling composition, district-heating grid presence∗, congestion flag∗, policy plan.
Technology (global)	investment cost∗, cumulative installs∗, lifetime, efficiency, energy source and price∗, subsidy, sustainability score, salience∗, learning rate.

Process overview and scheduling. Each simulated year executes the following schedule (Figure 2):

1. **Ageing and exogenous change.** All heating systems age by one year. A fraction of dwellings receive exogenous insulation and heat-distribution upgrades (independent of the heating decision).
2. **District-heating expansion.** The district heating company evaluates each neighborhood and starts (or continues) grid construction according to its strategy.

3. **Adoption decisions.** Actors decide in a fixed order. Social-housing and owners'-association blocks decide first, followed by landlords. Both are triggered when the heating system reaches the end of its life. Under the policy-based strategy, a social-housing block in a neighborhood planned for district heating also connects in the year the heat network arrives. Owner-occupiers decide last, triggered by the end of life or, past 75 % of the system's life, by an opportunity. Dwellings on district heating stay connected. Owner-occupiers are processed in two passes. The first computes the equivalent annual cost of every option for every triggered owner-occupier, so that the cost scale used to normalize affordability covers all decisions of that year. The second evaluates the theory-of-planned-behavior utility and makes a random-utility choice.
4. **Grid reaction.** The grid operator recomputes peak electricity demand per neighborhood and reinforces congested neighborhoods up to its annual rate.
5. **Global updates.** Technology salience and the economic learning curves are updated from the year's installations, feeding the next year's decisions.

State is updated asynchronously within a year (an agent's decision is visible to agents acting later that year), while learning and salience are updated synchronously at year-end.

Appendix A.1.2. Design concepts

Basic principles. Household adoption follows the theory of planned behavior (Ajzen, 1991), based on the application by Okur et al. (2024) embedded in a discrete random-utility choice. System-level change is driven by economic learning (Wright's law), social learning (a salience-mediated subjective norm), and threshold-based collective infrastructure decisions. These generate the transition dynamics described in Section 3.

Because the underpinning data are limited, uncertainty is large, and the analysis design treats each type of uncertainty separately (Table A.5). Stochastic variability arises at the level of individual households and largely averages out over the many dwellings in a run, so that few replications are required. The calibration and the national scenarios use two, the Morris screening one and the Latin-hypercube sample eight. Parameters that act in the 2022–2024 observation window mostly act on the same terms as the decision weights. For example, the attitude distribution and the perceived

sustainability of each heating method act on the attitude term. When the weights are free they compensate, so calibrating these parameters jointly would widen the trade-offs between them rather than constrain them. They are therefore held at their defaults in the calibration and varied afterwards in the sensitivity analysis. Only the scale of the random term is calibrated with the weights, because it scales the whole utility rather than one term of it.

Table A.5: Treatment of each class of uncertainty and the section in which the corresponding result is reported.

Class	**Origin in the model**	**Treatment**	**Reported in**
Stochastic	Random draws for attitudes, network formation, failure ages, random-utility term	Replications with independent random draws and averaged out over the dwellings in a run	Within the ensemble bands
Parametric, constrained by observations	The four decision weights and the noise scale	History matching against the observed 2022–2024 change (Section 6)	Section 6.1; ensemble bands in Section 7
Parametric, not constrained	Behavioral, structural and techno-economic parameters	Morris screening and Latin-hypercube sample at calibrated weights (Appendix C.1)	Section 7.1
Input (scenario)	Strength of the feedback loops, energy prices, policy measures and infrastructure	Explicit scenarios rather than sampled	Sections 7.2 and 7.4
Structural	Form of the decision function and form of the social-visibility curve	One alternative cost perception tested, the model itself grounded in the literature (Section 4)	Section 7.5; limitation (Section 8.1)

Emergence. Aggregate heating-mix pathways, including their accelerations, delays and lock-ins, emerge from individual decisions and their feedback mechanisms. Prices, costs and social norms that individual agents respond to are themselves outcomes of the population's decisions.

Adaptation. Triggered owner-occupiers adapt by selecting the option with the highest perceived utility among those technically available to them. Landlords and social housing associations adapt by selecting the lowest equivalent-annual-cost option. Infrastructure agents adapt their expansion to demand and policy.

Objectives. Owner-occupiers maximize a composite utility (affordability, ease, attitude, and social conformity). Landlords and associations minimize the annual cost. The district heating company implicitly pursues a positive business case, or follows policy. The grid operator maintains capacity.

Learning. Two reinforcing loops operate. Economic learning lowers a technology's investment cost as its cumulative installed base grows (Wright's law). Social learning raises a technology's salience, and thereby the subjective norm favoring it, as its adoption share grows. Both loops can turn a marginal advantage into self-sustaining growth.

Prediction. Agents are myopic, they do not forecast future prices, costs or adoption. Each decision uses the current year's costs and prices, held constant over the equipment's economic lifetime when computing the equivalent annual cost.

Sensing. Owner-occupiers sense the current heating methods of the peers in their social network and the availability of options (whether a district-heating grid exists in their neighborhood, and whether grid congestion blocks electric heat pumps). All agents sense the current technology costs and their own dwelling's characteristics.

Interaction. Direct interaction is limited to owner-occupiers observing network peers. Most interaction is indirect through the shared learning curve, through social norms, through the district-heating grid (a collective asset), and through grid congestion (a shared constraint). Social housing associations interact strongly with the collective actors by shifting whole neighborhoods at once.

Stochasticity. Stochasticity enters at initialization (household attitudes, peer-network formation, the assignment of ownership and current heating method from neighborhood statistics, and each system's end-of-life age, drawn from a normal distribution around its nominal lifetime) and at decision time (a

Gumbel random-utility term that lets choices deviate from the utility optimum). Results are therefore reported as means and spreads over replications and weight sets (Table A.5).

Collectives. Two collectives are represented explicitly. A social-housing block is the set of an association's dwellings within one postcode (roughly a street or complex) that decide and switch together. Home-owners' association blocks do the same for dwellings on collective gas heating. Neighborhoods are collectives for infrastructure, the district-heating grid and grid congestion are neighborhood-level states that all resident dwellings share.

Observation. For each scenario and year, the model records the installed stock and the annual installed/removed numbers of each heating method by ownership category, together with the mean of each decision term (attitude, subjective norm, perceived behavioral control, utility) and the mean equivalent annual cost. Derived outputs include technology-share pathways, their breakdown by segment, and the mix of cost-optimal, highest-utility and chosen options.

Appendix A.1.3. Details

Initialization. The dwelling stock is constructed from Dutch building administration and neighborhood statistics (Section 4). Each dwelling is assigned a construction year, type and floor area, an ownership category (owner-occupier, social housing or landlord), an initial heating method, insulation label, and heat-distribution system drawn from the observed neighborhood heating mix of the initialization year, and an end-of-life age for its current system drawn around the system's lifetime.

Ownership combines two resolutions. The neighborhood shares of owner-occupied, housing-corporation and other rental dwellings (CBS, 2026d) fix how many dwellings of each category a neighborhood contains, so that provincial totals reproduce the national housing-stock statistics (CBS, 2026e) to within half a percentage point. Postcode-level (PC6, about 20 dwellings) figures (CBS, 2026b) decide where they are located. Because housing corporations own whole streets and complexes rather than scattered dwellings, social housing is placed in clusters. Each postcode first receives its reported number of corporation dwellings, and the remainder of the neighborhood's social stock fills postcodes without a reported count one complete postcode at a time, in a random order weighted by their rental share. Postcodes without rental dwellings receive none. Private rental is then drawn per dwelling

from the remaining stock, weighted by the postcode's private-rental share, and the rest is owner-occupied. The social dwellings of each postcode form one housing block that decides collectively. Each block takes a single initial heating method from the neighborhood mix, with multi-family blocks served first because district and collective heating are concentrated in apartment buildings. Owner-occupied and privately rented dwellings on collective gas heating form owners'-association blocks that likewise decide collectively. Owner-occupier sustainability attitudes are drawn from a distribution parameterized from Okur et al. (2024) and Broers et al. (2019) and correlated with peer-network and insulation state through Cholesky decomposition (Edlin et al., 2015). Owner-occupiers are then connected in small-world networks that preferentially link households of similar attitude within the same municipality (Loomans et al., 2026). Table A.9 shows for 20 random owner-occupiers that the resulting networks link mainly households of the same adopter category. Technology salience is seeded from the initial stock shares, and cumulative installed bases are set to the initial stock so the learning curve starts from the present.

Input data. Building characteristics are from the Dutch cadastre (Kadaster, 2022); heat demand, insulation costs and energy-label transitions from VestaMAIS (PBL, 2017; van der Molen et al., 2021); ownership from the CBS neighborhood and postcode key figures (CBS, 2026d,b); observed per neighborhood heating mixes and their recent change from CBS; observed consumer energy prices from CBS (CBS, 2026c) and their trajectories from the national Climate and Energy Outlook (PBL, TNO, CBS & RIVM, 2026); district-heating tariffs from the ACM (ACM); and technology investment cost, lifetime, efficiency, subsidy and learning-rate assumptions as listed in Appendix Appendix A.2.

Submodels. Triggering. A decision is triggered when a system reaches the end of its life (problem-based) or, for owner-occupiers only, once its age exceeds 75 % of its life (opportunity-based). The trigger type governs whether an agent evaluates options, not the option costs directly. A dwelling on district heating has no end of life, it is reviewed every five years since connecting and only changes if another option is chosen. *Owner-occupier adoption* evaluates, for each available option, attitude, subjective norm and perceived behavioral control, combines them into an intention and a utility (Equations 1–4), and selects the highest-utility available option with an added Gumbel term (Equation 5). *Landlords, housing associations and owners' associations* select the

option with the lowest equivalent annual cost (Equation A.2, Appendix Appendix A.2), with a small Gumbel term. Blocks use the household-averaged cost and switch as a whole. Social housing associations may instead follow the neighborhood policy plan. *District-heating expansion* builds a grid in a neighborhood once the heat demand of interested dwellings crosses the positive-business-case threshold ($\sim$600 GJ/ha) or, under the policy strategy, where the heat-transition-vision plan specifies, after a construction lead time. *Grid reinforcement* recomputes neighborhood peak demand (including an exogenous electric-vehicle load, see Appendix A.4) and reinforces congested neighborhoods up to the annual reinforcement rate. *Economic learning* updates each technology's investment cost by Wright's law from its cumulative installations (Equation A.7), scaled by the scenario learning factor. *Salience*, which enters the subjective norm (Equation 1), is updated each year from the population share x_{hm} of method hm by an S-curve combining a novelty term (favoring low-share methods) and a momentum term (favoring methods whose share is rising):

$$\sigma_{hm} = m\,\nu + (1-m)\,x_{hm}, \qquad \nu = \frac{1}{1+e^{\,k(x_{hm}-\theta)}}, \qquad m = \frac{1}{1+e^{-\beta\,(x_{hm}-x_{hm,t-1})}} \tag{A.1}$$

where θ is the novelty threshold, k its steepness and β the momentum steepness. Finally, the affordability term in Equation 2 normalizes the equivalent annual cost on a logarithmic min–max scale spanning all methods and dwellings available in the year, so that a household's decision reflects the *proportional* cost differences between its options rather than being dominated by the most expensive dwellings. Scenario submodels (grid reinforcement rate, district-heating strategy, connection obligations and the congestion heat-pump ban) are detailed in Appendix A.3.

Appendix A.2. Cost calculations

Appendix A.2.1. EAC calculations

The costs are calculated using the Equivalent Annual Costs (EAC) for a fair comparison between heating methods with different investment costs, operational costs, and lifetimes (Hastings, 2021). Furthermore, the costs depend on the required insulation levels and in-house heat distribution system. For all-electric heat pumps, low-temperature in-house heat distribution and a high level of insulation is required for efficient operation. If these are not

yet installed, additional costs are added to the investment costs. This does of course lead to reduced heat demand, which decreases costs. These cost calculations are made for all dwellings when the adoption functions are triggered, independent of their owner.

The equivalent annual cost of option j is

$$EAC_j = \frac{1}{A_{T,r}} \left(I_j - S_j + \sum_{t=0}^{T-1} \frac{C_j}{(1+r)^t} \right), \qquad A_{T,r} = \frac{1-(1+r)^{-T}}{r}, \quad \text{(A.2)}$$

with I_j the investment, S_j the subsidy, C_j the annual running cost, T the lifetime and r the discount rate. The discount rate is 2 % over 15 years for the individual systems and 3 % over 30 years for a district-heating connection. The investment is

$$I_j = IC_{sys,j} + IC_{dist} + IC_{ins}. \quad \text{(A.3)}$$

Here $IC_{sys,j}$ is the cost of the system itself. It is a fixed price for the gas boiler, follows Equation A.5 for the heat pumps and is the connection fee for district heating. IC_{dist} and IC_{ins} are the costs of a low-temperature distribution system and of insulation to the required label (Equation A.6), added only when the option requires them and the dwelling lacks them. The annual running cost is

$$C_j = \sum_i \frac{f_{ij}\, Q}{\eta_{ij}}\, p_i + M_j + F_j, \quad \text{(A.4)}$$

with Q the heat demand of the dwelling, f_{ij} the share of heat from energy carrier i, η_{ij} its conversion efficiency, p_i its price, M_j the maintenance cost and F_j the fixed connection charges.

For district heating the terms differ (Table A.6): IC_{heat} is the connection fee for a dwelling that newly connects and zero for one that is already connected, and the fixed charge, metering fee and delivery-set rental are annual costs. A dwelling that leaves district heating adds the disconnection fee to the investment cost of its new system. The heat price is per kWh delivered, so no conversion efficiency applies.

The individual heating systems are parameterized from the cost figures that CE Delft compiled for the PBL model Vesta MAIS (CE Delft, 2023) (Table A.7). These are given excluding VAT at 2020 prices and are converted to constant 2024 euros including VAT with the consumer price index, as the energy prices are. The investment itself is set at the current consumer prices

Table A.6: District-heating costs, regulated maximum tariffs for a small connection, 2025, in euros including VAT (ACM).

Component	**€**	**Applies to**
Annual costs (€/yr)		
Fixed charge	577.48	Connected dwellings
Metering fee	32.80	Connected dwellings
Delivery-set rental	150.49	Connected dwellings
One-off costs (€)		
Connection fee	5 251.45	New connections, minus subsidy
Disconnection fee	5 250.99	Dwellings that leave

of Milieu Centraal, because the CE Delft figures, a 2020 cost price indexed with general inflation whose upper bound is a standalone replacement, lie 25–50 % above them for heat pumps. A gas boiler costs 2 100 EUR including installation (Milieu Centraal, 2026a) whatever the heat demand, because a combi boiler is chosen by its hot-water class rather than its heating capacity. The heat pumps are priced by capacity, because their cost rises with the capacity a dwelling needs:

$$IC_{hp} = IC_0 + c_{kW}\,P, \qquad P = \min\left(P_{max},\, \max\left(P_{min},\, s\,\frac{Q}{h_{fl}}\right)\right) \tag{A.5}$$

with Q the dwelling's heat demand, h_{fl} the full-load hours, s the share of the peak covered by the heat pump (1 for the electric heat pump, 0.5 for the hybrid, whose capacity is limited to 2–6 kW) and c_{kW} the cost per kW used in Vesta MAIS 5.0 (410 EUR/kW for an air-to-water heat pump, the midpoint of 320–500, and 250 EUR/kW for a hybrid in 2020 prices excluding VAT, converted as above). The level follows Milieu Centraal, including VAT and installation: 12 000 EUR for an 8 kW electric heat pump in a terraced house (Milieu Centraal, 2026c) and 8 300 EUR for a 4 kW hybrid with a new boiler (Milieu Centraal, 2026b), The hybrid includes the boiler because the model treats it as one system with one lifetime. $h_{fl} = 1\,220$ hours gives the median terraced house in the stock (9 772 kWh) the 8 kW of the reference price. The CE Delft figures bound the sensitivity range. Both heat pumps use one seasonal performance factor for space heating and hot water together, because the model has a single heat demand per dwelling, and the hybrid delivers 80 % of the heat from its heat pump. Both heat pumps learn at 10 % per doubling of their cumulative installations (Louwen et al., 2018).

In the sensitivity analysis we vary this between zero, as observed for recent installed costs in the United Kingdom (Renaldi et al., 2021), and the 28 % found historically for heat-pump prices (Weiss et al., 2010).

Table A.7: Heating-system assumptions, constant 2024 euros including VAT. Heat-pump investment follows Eq. (A.5) and is shown for heat demands of 5 500 / 9 772 / 17 400 kWh (about the median apartment, terraced and detached house). Sources: CE Delft (2023) and, for the investment, Milieu Centraal

	Gas boiler	Hybrid heat pump	Electric heat pump	District heating
Investment (€)	2 100	7 661 / 8 302 / 9 032	9 901 / 12 006 / 15 764	5 251
Low-temperature distribution (€)	–	3 063	3 063	–
Maintenance (€/yr)	203	73	73	–
Lifetime (years)	15	15	15	–
Efficiency	0.94	3.8 / 0.94	3.8	1.0
Learning rate (per doubling)	1 %	10 %	10 %	–

Insulation costs and benefits in heat demand reduction are based on a table with, for each dwelling type and building year category, insulation costs from label x to y, calculated with a fixed and a variable per-m^2 part as follows:

$$IC_{ins} = IC_{fixed} + IC_{var} \times A_{ins} \tag{A.6}$$

Where:

- IC_{ins} = Total insulation cost
- IC_{fixed} = Fixed insulation cost
- IC_{var} = Variable insulation cost per m^2
- A_{ins} = Insulated area (m^2)

Appendix A.2.2. Economic learning

The new cost per unit after additional installations is calculated using the learning curve formula (Wright's law) for each heating method option (Equation A.7). Note, in this we assume that the learning rate in the Netherlands

specifically is similar to the global learning rate in global markets, as we consider installed units on a national level.

$$C_n = C_{init} \times \left(\frac{Q_n}{Q_0}\right)^{\log_2(1-LR)} \tag{A.7}$$

Where:

- C_n = New cost per unit
- C_{init} = Initial cost per unit
- Q_{init} = Initial number of units installed
- Q_n = New total number of units installed ($Q_n = Q_0 + Q_{add}$)
- LR = Learning rate (cost reduction per doubling of cumulative production)

Appendix A.2.3. Subsidies, renovation and equipment lifetime

Investment subsidies lower the up-front cost of heat pumps and heat-network connections. They follow the Dutch schemes year by year. For owner-occupiers and owner associations the national investment subsidy for sustainable energy (ISDE), which for a typical hybrid heat pump was 2 550 euros in 2022–2024 and 1 925 euros in 2025, for an all-electric heat pump 3 150 and 2 825 euros, and for a heat-network connection 3 325 euros until 2023 and 3 775 euros from 2024 (RVO, 2026). For private landlords the corresponding scheme for rental dwellings. For housing associations a contribution to heat-network connections only. After 2025 the amounts are held at their 2025 level. The calibration therefore sees the subsidies that applied over 2022–2024, and the scenarios vary them (Appendix A.3).

The building stock is renovated approximately once every 25 years, independently of heating decisions. These renovations are exogenous to the model, but because they change the cost comparison, insulation levels and heat distribution systems are improved at random in dwellings every 25 years. Equipment does not fail deterministically. Each installed system's end-of-life age is drawn from a normal distribution around its nominal lifetime (standard deviation 1 year, limited to ±3 years), so that replacements are spread out. A dwelling on district heating has no end of life of its own, it stays connected unless a scenario lets it review its heating.

Appendix A.3. Scenario input

The scenarios are created by varying several parameters per actor. The exact description of these parameters and possible options is given in Table A.8.

Appendix A.4. Grid impact

Peak electricity demand per neighborhood includes an exogenous load from electric vehicles. Its growth follows an S-curve scaled to the number of cars in the neighborhood and their 2023 share of electric vehicles (Figure A.18).

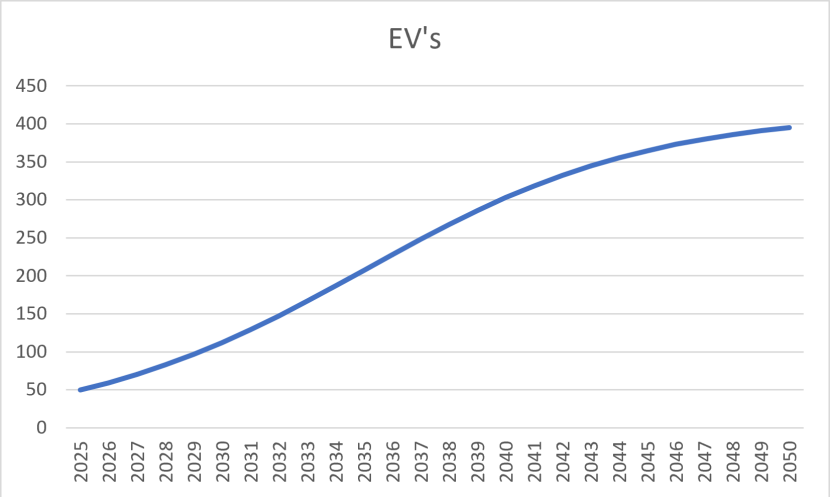


Figure A.18: S-curve EV adoption for a random neighbourhood. The curve is similar for all neighbourhoods based on the total number of cars and the 2023 share of EVs.

Table A.8: Strategies per actor and policy measures tested in the scenario analysis. The grid reinforcement rate (medium) and the construction time of heat networks (5 years) are the same in all scenarios.

Actor	**Parameter**	**Options**	**Description**
Household	Social learning factor	0.5, 1, 2	Multiplies the weight of the subjective norm in the intention.
Social housing association	Heating strategy	COST_BASED, POLICY_BASED	A block chooses the option with the lowest equivalent annual cost or follows the neighborhood heat plan.
District heating operator	Expansion strategy	COST_BASED, POLICY_BASED	Networks are built where the heat demand of interested households makes them profitable, or where the heat plan specifies.
–	Energy prices	central, gas high and electricity low, gas low and electricity high	Price paths of gas and electricity.
–	District heating connection obligation	FALSE, TRUE	Households in neighborhoods with a heat network must connect to it.
–	Congestion heat pump ban	FALSE, TRUE	Electric heat pumps are banned in neighborhoods with grid congestion.
–	Economic learning factor	0.5, 1, 2	Multiplies the learning rate of every technology, so that heat pumps learn at 5, 10 or 20 % per doubling of cumulative installations.
National government	Investment subsidy	CURRENT, NONE, PHASE_OUT	Subsidies for heat pumps and heat-network connections held at their 2025 level, no subsidy, or the current subsidy until the technology reaches 20 % of the housing stock.

Table A.9: Network similarity and sizes for 20 random owner-occupiers, used to verify the network algorithm.

Owner-occupier	Owner attitude	Owner category	Network attitude	Network size	Same category
0	0.687	4	0.660	19	79 %
1	0.923	2	0.877	22	86 %
2	0.686	4	0.655	25	96 %
3	0.933	2	0.882	29	90 %
4	0.763	3	0.769	28	89 %
5	0.665	4	0.655	23	91 %
6	0.506	5	0.569	25	76 %
7	0.829	3	0.777	24	83 %
8	0.587	4	0.684	24	75 %
9	0.480	5	0.518	24	92 %
10	0.516	5	0.556	26	81 %
11	0.211	6	0.343	25	96 %
12	0.757	3	0.792	27	89 %
13	0.687	4	0.642	30	83 %
14	0.712	4	0.681	27	78 %
15	0.984	2	0.882	26	89 %
16	0.746	3	0.779	32	88 %
17	0.683	4	0.664	31	71 %
18	0.859	3	0.801	22	91 %
19	0.861	3	0.794	24	92 %

Appendix B. Calibration details

Appendix B.1. Calibration criterion

A weight set is judged by the implausibility of each scored output k (technology and year), the standard measure in history matching (Vernon et al., 2010; Andrianakis et al., 2015):

$$I_k = \frac{|s_k - o_k|}{\sqrt{\sigma^2_{\mathrm{MC},k} + \sigma^2_{\mathrm{obs},k} + \sigma^2_{\mathrm{disc}}}}, \tag{B.1}$$

where s_k is the simulated share and o_k the observed share in percentage points. The denominator collects the three reasons why a plausible weight set would not match the observation exactly: the stochastic variation of the model (σ_{MC}, at most 0.06 percentage points), the uncertainty of the observation ($\sigma_{\mathrm{obs}} \approx 0.29$ percentage points). This mainly reflects the 1.8 % of

dwellings whose heating type is not allocated in the published neighborhood data. It is estimated by allocating this share at random over the technologies, the same way in every neighborhood, and by adding the rounding error of the published shares, which contributes little. And third, the model discrepancy (σ_{disc}), an allowance for the model being a simplified account of the heat transition, set to 0.5 percentage points. A set is retained when the implausibility (Equation B.1) stays below 3 for every scored output, the conventional cutoff that follows from the three-sigma rule (Pukelsheim, 1994). This admits a deviation of about 1.7 percentage points on every scored output. Because σ_{disc} and the cutoff are judgments, Table B.10 reports how many sets are retained when either is varied.

Each weight set is run for two iterations. This is sufficient, because the standard deviation between iterations, at most 0.06 percentage points, is small compared with the deviation of 1.7 percentage points that the criterion admits.

Table B.10: Number of the 1000 sampled weight sets retained, by model-discrepancy allowance σ_{disc} and implausibility cutoff. The central choice is in bold.

σ_{disc} (pp)	$I < 2.5$	$I < 3.0$	$I < 3.5$
0.25	33	60	83
0.50	87	**124**	157
1.00	255	675	753

Appendix B.2. Selecting the subsample

Running every retained set through every scenario on a national scale is not feasible, so the scenario analysis uses a subsample of 25 of the 124 retained sets. Because the results are reported as the median and percentiles across the sets, the subsample is chosen so that these statistics reproduce those of the full retained ensemble. To this end, the baseline is run for every retained set in Noord-Brabant, and the subsample is the combination of 25 sets whose 5, 25, 50, 75 and 95 % percentiles of the share of each technology in 2030, 2040 and 2050 come closest to those of all retained sets (the smallest sum of squared differences, standardized by the spread over the retained sets, found by a local search with random restarts). The chosen sets count equally. In the selection years, the subsample reproduces the medians of the full ensemble within 1.1 and the 5 and 95 % percentiles within 1.3 percentage

points. The bands in the scenario analysis are therefore the spread across the plausible weight sets, with the median as a descriptive center, and not probability intervals.

The sensitivity analysis takes its weight sets from the same subsample. The Morris screening is repeated at four sets that span the two ways the retained sets differ: the best-fitting set, the two ends of the cost–attitude line (the most cost-driven and the most attitude-driven retained set), and the set with the highest effective weight of the subjective norm. The Latin-hypercube quantification is run at the single best-fitting set so that all single-set results refer to one and the same point in the ensemble.

Appendix C. Sensitivity assumptions

Appendix C.1. Parameters of the sensitivity analysis

Table C.11 lists the parameters screened with the Morris method, with the range over which each is varied, its default value in the scenario runs, and the basis of the range. The ranges of the cost parameters are taken from the literature. Most behavioral and structural ranges are modelling choices, because data on these aspects of the decision are lacking. Two parameters that also shape the decision are not screened. The noise scale of the choice λ is calibrated together with the decision weights (Section 6), and the effort of switching enters the decision only through its product with the calibrated effort weight, so that varying it would count the same uncertainty twice. The review interval and exit barrier of district heating are not screened either, because connected dwellings stay connected in all scenarios.

The Morris method (Morris, 1991; Campolongo et al., 2007) moves one parameter at a time along random trajectories through the parameter space. Each step changes a single parameter and records the resulting change in the outcome, an elementary effect. The mean absolute elementary effect, μ^*, ranks the parameters by influence, the standard deviation of the elementary effects, σ, indicates whether a parameter acts on its own or in combination with others. The screening uses eight trajectories on a four-level grid for each of the four screening weight sets (Section 6.2), in the baseline scenario for Noord-Brabant over 2024–2050, with one iteration per run. Each parameter is scaled to its range, so that μ^* is the change in a technology's 2050 share, in percentage points, for a move across the full range. The effects are averaged over the four sets, cost parameters weigh more in the cost-driven sets, and attitude and norm parameters in the attitude-driven sets.

The seven highest-ranked parameters (marked †) are then varied jointly in a Latin-hypercube sample of 80 combinations, each drawn uniformly over the ranges in Table C.11 and run with eight iterations, at the best-fitting weight set, in the same scenario, region and period. All other parameters are held at their defaults. For each technology's 2050 share we report the spread across the 80 combinations and, from a linear regression on the parameters, the change in the share when a parameter moves across its range. The share of the variation that this linear model explains (leave-one-out R^2), compared with a model that adds squared and pairwise interaction terms, shows how far the parameters act independently of each other.

Table C.11: Parameters of the sensitivity analysis, grouped by the part of the decision function they enter. Range: low–high (default in the scenario runs). †: also varied in the Latin-hypercube sample.

Parameter	**Range (default)**	**Meaning**	**Basis**
Attitude			
Attitude distribution†	α 3–7 (5)	Attitudes drawn from Beta($\alpha, 2$), mean 0.60–0.78 (0.71)	
Perceived sustainability of the hybrid†	2–4 (3)	Position on the 1–5 scale (gas 1, electric heat pump 5)	Around the data value
Subjective norm			
Social learning strength†	0.5–2 (1)	Multiplier on the weight of the subjective norm	Scenario low and high values
Peer network size†	10–50 (25)	Peers per household	
Peer network homophily	0.5–0.95 (0.8)	Share of peers similar to the household	
Salience: novelty threshold	0.15–0.45 (0.3)	Population share below which a method counts as novel	
Salience: novelty steepness	5–20 (10)	Steepness of the novelty term	
Salience: momentum steepness	10–50 (30)	Steepness of the momentum term	
Costs (affordability)			
Economic learning rate†	0–28 % (10 %)	Fall in investment per doubling of cumulative installations	0 % (Renaldi et al., 2021), 28 % (Weiss et al., 2010)
Heat-pump capital cost†	0.8–1.3 (1)	Multiplier on heat-pump investment	0.8 market prices a fifth lower, 1.3 midpoint of CE Delft (2023)
Energy price path	low–high (central)	Gas and electricity prices between the low, central and high arm	PBL, TNO, CBS & RIVM (2026)
District-heating connection cost	0.85–1.15 (1)	Multiplier on the regulated connection fee	±15 % around ACM
Randomness in perceived costs	€ 5–50 (20)	Random term in the cost-only choice of landlords and housing associations	
Structural			
Opportunity trigger†	0.6–0.9 (0.75)	Share of a system's lifetime after which it may be replaced before failure	
Equipment lifetime spread	0–3 (1) years	Standard deviation of the end-of-life age, limited to ±3 years	